

# **Enhancing Learning Style Classification Using Random Forest-Based Semi-Supervised Labeling**

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**A Capstone project report submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science and Engineering**



**Department of Computer Science and Engineering**

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**Dhaka-1212, Bangladesh**

**Date: 15 May, 2025**

## Declaration

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We, **Sanjida Alam Lamia, Abidur Rahman, Farhan Tonoy, and Golpo Dev**, hereby declare that the work presented in this capstone project report is the outcome of the investigation performed by us under the supervision of **Dr. Md. Atiqur Rahman**, Assistant Professor, Department of Computer Science and Engineering, East West University. We also declare that no part of this project has been or is being submitted elsewhere for the award of any degree or diploma, except for publication.

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## Letter of Acceptance

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The capstone project report entitled **Enhancing Learning Style Classification through Random Forest-Based Semi-Supervised Labeling**, submitted by Sanjida Alam Lamia, Abidur Rahman, Farhan Tonoy, and Golpo Dev, to the Department of Computer Science and Engineering, East West University, Dhaka, Bangladesh, is accepted for partial fulfillment of the requirement for the degree of Bachelor of Science in Computer Science and Engineering on May 15, 2025.

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## Abstract

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In the era of personalized education, adapting learning content to individual learner preferences is crucial for improving engagement and academic outcomes. Learning styles play a vital role in this adaptation, and the Felder-Silverman Learning Style Model (FSLSM) provides a well-structured framework for classifying learners across four dimensions: Active–Reflective, Sensing–Intuitive, Visual–Verbal, and Sequential–Global. Traditionally, learning styles are identified using questionnaires like the Index of Learning Styles (ILS), which are time-consuming, subjective, and prone to inconsistencies. This project focuses on developing a machine learning-based system for the automatic classification of learning styles based on students’ behavioral data in e-learning platforms, eliminating the need for manual questionnaires. Our project began with an in-depth literature review, identifying key methodologies and limitations in existing work. We selected FSLSM as the core model and extracted behavioral features aligned with its four dimensions. Two datasets were used: one from CSMS (Computer Skills for Medical Students), containing 1952 data instances, and another from CSHS (Computer Skills for Humanities Students), containing 670 data instances. These datasets include clickstream-based features such as time spent, content navigation, quiz behavior, and resource preferences, which reflect user interaction patterns. To ensure a balanced training environment, the Synthetic Minority Oversampling Technique (SMOTE) was applied to address class imbalance across FSLSM dimensions. Several machine learning models were implemented and tested, including XGBoost, Decision Tree, K-Nearest Neighbors (KNN), and Support Vector Machine (SVM). Among them, Random Forest achieved the best performance, with an accuracy of 94.33% on the CSMS dataset and 93% on the CSHS dataset, outperforming other models in terms of precision, recall, and F1-score. These results validate the effectiveness of Random Forest in handling non-linear feature interactions and imbalanced class distributions. Future work will focus on expanding the dataset with data from diverse learner populations across various academic disciplines to improve generalizability. We also plan to integrate explainable AI (XAI) techniques such as SHAP or LIME to provide interpretability of model predictions, making the system more transparent to educators and learners. Furthermore, we aim to incorporate real-time data streams from learning management systems (LMS) for dynamic learning style prediction and adaptation. The inclusion of deep learning models such as LSTM or Transformers may enhance pattern recognition in sequential learning behavior. Lastly, developing a user-friendly dashboard or plugin for integration with existing e-learning platforms could facilitate real-world deployment and practical usage of the system by instructors and educational institutions.

## Acknowledgments

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As is true for everyone, we have also arrived at this point of achieving a goal in our life through various interactions with and help from other people. However, written words are often elusive and harbor diverse interpretations even in one's mother language. Therefore, we would not like to make efforts to find the best words to express our thankfulness other than simply listing those people who have effectively contributed to this thesis itself. This work was carried out in the Department of Computer Science and Engineering at East West University, Bangladesh.

First, we would like to express our deepest gratitude to the Almighty for His blessings on us. Next, our special thanks go to our supervisor, Dr. Md. Atiqur Rahman, who gave us this opportunity, initiated us into the field of "Machine Learning", and without whom this work would not have been possible. His encouragement, vision, thoughtful comments and suggestions, and unforgettable support at every stage of our B.Sc. study were simply appreciative and essential. His ability to muddle us enough to finally answer our question correctly is something valuable that we have learned, and we will try to emulate it if we ever get the opportunity.

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**Sanjida Alam Lamia**  
15 May, 2025

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15 May, 2025

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15 May, 2025

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15 May, 2025

# Contents

|                                                      |           |
|------------------------------------------------------|-----------|
| Declaration . . . . .                                | 2         |
| Letter of Acceptance . . . . .                       | 3         |
| Abstract . . . . .                                   | 4         |
| Acknowledgments . . . . .                            | 5         |
| <b>List of Figures</b>                               | <b>8</b>  |
| <b>List of Tables</b>                                | <b>9</b>  |
| <b>List of Acronyms</b>                              | <b>10</b> |
| <b>1 Introduction</b>                                | <b>11</b> |
| 1.1 Background and Motivation . . . . .              | 11        |
| 1.2 Problem Statement and Analysis . . . . .         | 12        |
| 1.3 Project Objectives . . . . .                     | 13        |
| 1.4 Project Contributions . . . . .                  | 13        |
| 1.5 Project Outline . . . . .                        | 14        |
| <b>2 Literature Review</b>                           | <b>15</b> |
| 2.1 Survey of the State-of-the-art . . . . .         | 15        |
| 2.1.1 Summary . . . . .                              | 20        |
| <b>3 Materials and Method</b>                        | <b>23</b> |
| 3.1 Materials . . . . .                              | 23        |
| 3.1.1 Dataset Collection . . . . .                   | 23        |
| 3.1.2 Dataset Exploration . . . . .                  | 23        |
| 3.1.3 Dataset Sampling . . . . .                     | 24        |
| 3.1.4 Dataset Processing . . . . .                   | 24        |
| 3.1.5 Research Environment and Devices . . . . .     | 24        |
| 3.2 Methods . . . . .                                | 25        |
| 3.2.1 Proposed Models . . . . .                      | 25        |
| 3.2.2 Design/Framework . . . . .                     | 25        |
| 3.2.3 Algorithms/Models Formulation . . . . .        | 26        |
| 3.2.4 Experimental Setup . . . . .                   | 29        |
| 3.3 Implementation . . . . .                         | 29        |
| 3.4 Summary . . . . .                                | 29        |
| <b>4 Results &amp; Discussion</b>                    | <b>31</b> |
| 4.1 Obtained Results . . . . .                       | 31        |
| 4.2 In-depth Result Analysis . . . . .               | 31        |
| <b>5 Conclusion</b>                                  | <b>51</b> |
| 5.1 Overall Contributions . . . . .                  | 51        |
| 5.2 Limitations . . . . .                            | 51        |
| 5.3 Future Work . . . . .                            | 51        |
| <b>Bibliography</b>                                  | <b>53</b> |
| 5.4 Mapping of Course and Program Outcomes . . . . . | 55        |
| 5.4.1 CSE400-A . . . . .                             | 55        |

|       |          |    |
|-------|----------|----|
| 5.4.2 | CSE400-B | 57 |
| 5.4.3 | CSE400-C | 58 |

# List of Figures

|      |                                                                                        |    |
|------|----------------------------------------------------------------------------------------|----|
| 1.1  | The Felder-Silverman Learning Style Model . . . . .                                    | 12 |
| 3.1  | Overview of the Proposed Method . . . . .                                              | 26 |
| 4.1  | Confusion Matrix Analysis: RF Performance on Processing Target . . . . .               | 32 |
| 4.2  | ROC Curve for "Processing" Prediction Using RF Classifier . . . . .                    | 33 |
| 4.3  | Confusion Matrix Analysis: DT Performance on Perception Target . . . . .               | 35 |
| 4.4  | ROC Curve for "Perception" Prediction Using DT Classifier . . . . .                    | 36 |
| 4.5  | Confusion Matrix Analysis: Random Forest Performance on Input Target . . . . .         | 37 |
| 4.6  | ROC Curve for "Input" Prediction Using RandomForest Classifier . . . . .               | 38 |
| 4.7  | Confusion Matrix Analysis: Random Forest Performance on Understanding Target . . . . . | 40 |
| 4.8  | Enter Caption . . . . .                                                                | 40 |
| 4.9  | Confusion Matrix Analysis: Random Forest Performance on Processing Target . . . . .    | 42 |
| 4.10 | ROC Curve for "Processing" Prediction Using RandomForest Classifier . . . . .          | 43 |
| 4.11 | Confusion Matrix Analysis: Random Forest Performance on Perception Target . . . . .    | 45 |
| 4.12 | Enter Caption . . . . .                                                                | 46 |
| 4.13 | Confusion Matrix Analysis: Random Forest Performance on Input Target . . . . .         | 48 |
| 4.14 | Enter Caption . . . . .                                                                | 49 |



# List of Tables

|      |                                                                                                                                 |    |
|------|---------------------------------------------------------------------------------------------------------------------------------|----|
| 2.1  | List of the Related Works . . . . .                                                                                             | 20 |
| 4.1  | Supervised Learning Performance Comparison (Processing Target - Labeled Data Only) . . . . .                                    | 31 |
| 4.2  | Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing . .                                  | 32 |
| 4.3  | Supervised Learning Performance Comparison (Perception Target - Labeled Data Only) . . . . .                                    | 34 |
| 4.4  | Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing<br>(Perception Target) . . . . .     | 34 |
| 4.5  | Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing<br>(Additional Evaluation) . . . . . | 37 |
| 4.6  | Supervised Learning Performance Comparison (Understanding Target - Labeled Data Only) . . . . .                                 | 39 |
| 4.7  | Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing . .                                  | 39 |
| 4.8  | Supervised Learning Performance Comparison (Processing Target - Labeled Data Only) . . . . .                                    | 41 |
| 4.9  | Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing . .                                  | 41 |
| 4.10 | Supervised Learning Performance Comparison (Perception Target - Labeled Data Only) . . . . .                                    | 44 |
| 4.11 | Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing . .                                  | 44 |
| 4.12 | Supervised Learning Performance Comparison (Processing Target - Labeled Data Only) . . . . .                                    | 47 |
| 4.13 | Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing . .                                  | 47 |

## List of Acronyms

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|         |                                         |
|---------|-----------------------------------------|
| ALS     | Adaptive Learning System                |
| ANN     | Artificial Neural Network               |
| CNN     | Convolutional Neural Network            |
| CSHS    | Computer Skills for Humanities Students |
| CSMS    | Computer Skills for Medical Students    |
| DT      | Decision Tree                           |
| EDM     | Educational Data Mining                 |
| FCM     | Fuzzy C-Means                           |
| FSLSM   | Felder-Silverman Learning Style Model   |
| kNN     | k-Nearest Neighbors                     |
| LMS     | Learning Management System              |
| ML      | Machine Learning                        |
| SVM     | Support Vector Machine                  |
| SSL     | Semi-Supervised Learning                |
| XGBoost | Extreme Gradient Boosting               |

# CHAPTER 1: Introduction

---

Natural and human-made disasters like pandemics, wars, and natural events (earthquakes, floods, fires, etc.) have disrupted education worldwide. These events have pushed schools to shift from traditional classrooms to online learning. Tools like Learning Management Systems (LMS) have helped by offering features for online teaching, such as content sharing, communication, and progress tracking. However, LMS platforms often treat all students the same, ignoring their different learning styles. Learning styles are the different ways people prefer to learn. Traditionally, these styles are identified using questionnaires, but these can be inaccurate if students give random or incomplete answers. They also can't easily adapt if a student's learning style changes. Recently, machine learning has become popular in education because it can analyze large amounts of student data to understand learning behaviors. Using machine learning to detect learning styles can provide more accurate, flexible, and adaptive results than traditional methods. This leads to a more personalized learning experience for students. In this study, a self-taught machine learning method is used to identify students' learning styles. Although this approach shows promise, it must also align with how courses are designed. Course structure and missing data (like forum activity) can affect the model. To handle this, a semi-supervised learning method is used, which works well even with limited labeled data, helping to create a more adaptive and student-focused learning environment.

## 1.1 Background and Motivation

In the age of digital transformation, education has undergone a significant paradigm shift. The traditional, instructor-centered model of teaching has evolved into more dynamic, learner-centric ecosystems facilitated by Learning Management Systems (LMS). These platforms serve as the backbone of e-learning environments, offering tools for content delivery, student interaction, performance assessment, and data collection. However, while LMS platforms are technologically robust, they often lack intelligent mechanisms for personalizing the learning experience. One critical missing link is the ability to automatically identify and adapt to students' learning styles, which play a vital role in determining how effectively a learner assimilates knowledge.

Learning styles refer to the characteristic ways in which individuals prefer to receive, process, and retain information. Models like Felder-Silverman and Kolb's Experiential Learning Theory categorize students into distinct types (e.g., visual/verbal, active/reflective, sensing/intuitive). Numerous studies confirm that aligning teaching methods with these preferences enhances cognitive performance, engagement, and satisfaction. Traditionally, learning style classification has relied on self-assessment questionnaires. While these are easy to administer, they often suffer from subjective biases, inconsistency over time, and lack of scalability—especially in online learning settings where the student population is large and diverse.

To overcome these limitations, data-driven approaches are gaining popularity. Behavioral traces logged by LMS systems—such as clickstream data, quiz attempts, video watching habits, and resource downloads—provide rich indicators of student behavior. These data can potentially reveal learning preferences without direct questioning. However, using such data effectively poses significant challenges, especially due to the high prevalence of unlabeled data. Most students' learning styles remain unknown, and manually labeling them is impractical. This necessitates the use of semi-supervised learning (SSL) techniques, which can exploit both labeled and unlabeled data to improve classification performance. Prior research has explored this direction, particularly through self-training models. Nonetheless, these methods often suffer from instability, error propagation, and limited adaptability—gaps that this study aims to address through a new, more robust semi-supervised framework.

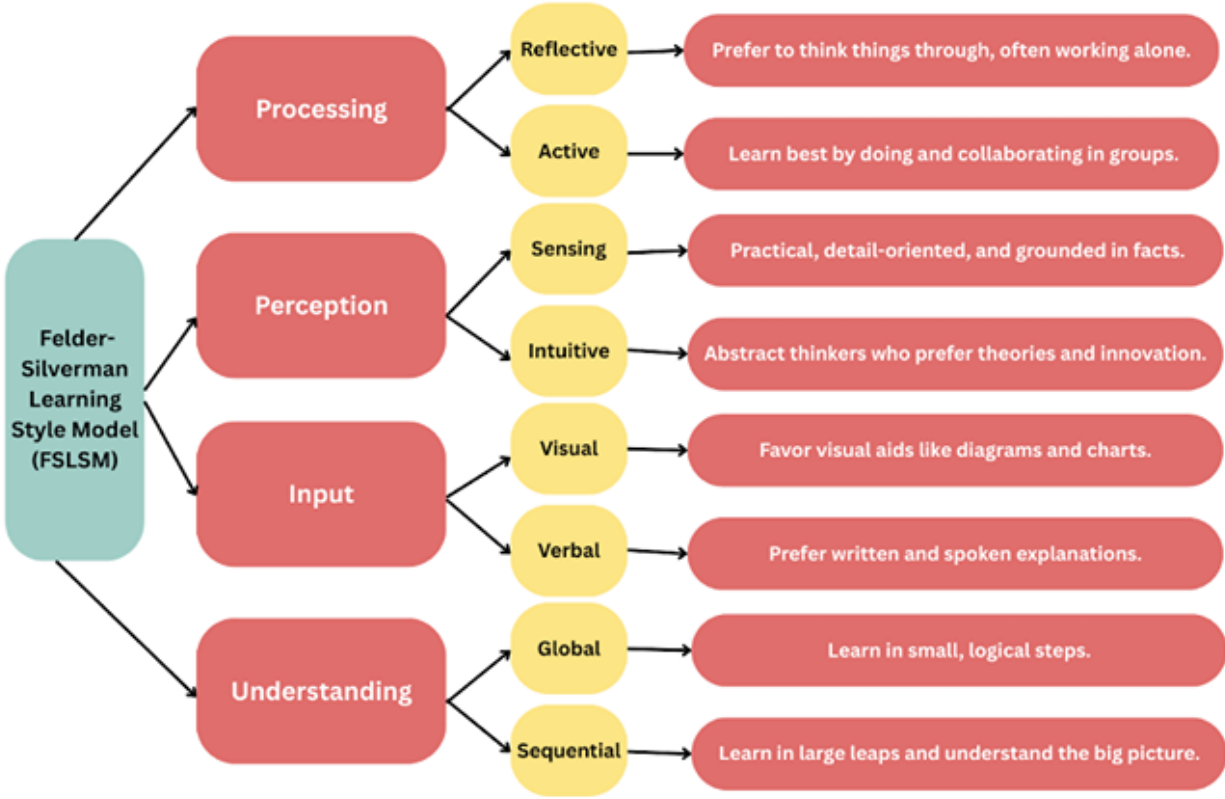


Figure 1.1: The Felder-Silverman Learning Style Model

## 1.2 Problem Statement and Analysis

While semi-supervised learning has opened promising avenues for automatic learning style detection, several key issues remain unresolved. The baseline self-training model proposed in prior work suffers from a reliance on high-confidence predictions to label unlabeled data iteratively. This can be problematic, as early classification errors often get reinforced in subsequent iterations, leading to cascading label noise and decreased model accuracy. Moreover, these models often operate on a single-layer preprocessing pipeline, where feature encoding, noise handling, and data balancing are not systematically optimized. In real-world educational datasets, such practices can significantly degrade the integrity of the learning signals.

Another limitation lies in the lack of statistical validation of features. Not all attributes in a student behavior dataset contribute equally to predicting learning styles. Some features might be noisy, irrelevant, or even misleading. Previous works seldom assess the statistical significance of each feature before feeding them into the model, resulting in feature redundancy and model overfitting. Additionally, many learning datasets are imbalanced, with certain learning styles being underrepresented. If left unaddressed, this imbalance causes bias in classification and limits the model's ability to generalize.

Lastly, there's a methodological gap in how labeled and unlabeled datasets are prepared and utilized. Most studies treat them as a monolithic unit, ignoring the fact that each dataset may have different missing patterns, feature distributions, or encoding needs. The lack of specialized handling for labeled versus unlabeled data further undermines model performance and adaptability. Addressing these weaknesses requires a more granular, multi-stage processing pipeline, which integrates advanced preprocessing techniques, robust statistical filtering, and intelligent semi-supervised modeling.

## 1.3 Project Objectives

The primary objective of this research is to design and implement a novel semi-supervised model for learning style identification that outperforms the baseline self-training method reported in the existing literature. To ensure a fair and direct comparison, the proposed model is applied to the same datasets used in the prior comparative study.

Our work addresses the following concrete goals:

1. **Framework Development:** Develop a robust semi-supervised framework capable of accurately labeling previously unlabeled data, minimizing label noise, and generalizing across diverse student populations.
2. **Dataset-Specific Preprocessing:** Independently preprocess the labeled (CSHS) and unlabeled (CSMS) datasets by applying missing value imputation and encoding strategies tailored to the unique characteristics of each dataset.
3. **Data Enrichment:** Use the proposed semi-supervised model to label the unlabeled CSMS data. Subsequently, merge the newly labeled CSMS data with the original CSHS dataset to form an enriched and comprehensive training dataset.
4. **Statistical Feature Filtering:** Apply hypothesis testing ( $p$ -value  $\leq 0.05$ ) on the training set to statistically eliminate non-contributing features. The same features are removed from the test set to maintain feature consistency.
5. **Class Imbalance Mitigation:** Experiment with synthetic data augmentation methods including SMOTE, SMOTE-Tomek, and SMOTE-ENN to address class imbalance within the training dataset.
6. **Feature Engineering:** Perform separate feature engineering and selection processes for the training and test sets to preserve their distinct properties while aligning the final feature vectors.
7. **Feature Normalization:** Normalize the selected features across both training and test sets to prepare them for model ingestion.
8. **Model Evaluation:** Evaluate the final classifiers using twelve diverse performance metrics: accuracy, precision, recall, specificity, AUC-ROC, Cohen's Kappa, log loss, and various time-based metrics.

This comprehensive pipeline ensures that every stage—from preprocessing to evaluation—is optimized for educational context sensitivity, feature quality, and overall performance accountability.

## 1.4 Project Contributions

This research offers several unique and impactful contributions to the field of Educational Data Mining (EDM) and adaptive e-learning systems:

1. **Novel Semi-Supervised Learning Model:** We propose a new semi-supervised learning approach that enhances traditional self-training by incorporating dynamic confidence thresholds, feature-aware labeling, and multi-round verification for error correction during pseudo-label generation.
2. **Dual-Dataset Methodology:** The study introduces a dual-dataset preprocessing pipeline, where labeled (CSHS) and unlabeled (CSMS) student data—originating from distinct academic domains—are treated separately. This promotes a more realistic and generalizable model architecture.
3. **Statistical Feature Selection:** Hypothesis testing is employed as a formal feature selection method. Features with  $p$ -values greater than 0.05 are filtered out early, improving interpretability and reducing dimensional noise.
4. **Advanced Resampling Techniques:** To combat class imbalance, the research integrates advanced data resampling strategies, including SMOTE and its variants (SMOTE-Tomek and SMOTE-ENN), which are particularly beneficial for underrepresented learning style classes.
5. **Multi-Metric Evaluation Framework:** A comprehensive performance evaluation is conducted using twelve diverse metrics—covering accuracy, sensitivity, specificity, calibration (log loss), agreement (Cohen's Kappa), and others—ensuring both statistical rigor and practical relevance.

6. **Reproducible Experimental Design:** The study ensures methodological reproducibility through a clearly defined pipeline, including strict separation of training and test datasets and fair benchmarking against baseline models from existing literature.

Collectively, these contributions set a new benchmark in automated learning style classification and lay the foundation for more intelligent, responsive Learning Management Systems (LMS) capable of personalizing education based on inferred cognitive profiles.

## 1.5 Project Outline

The structure of this paper is organized as follows:

- **Section 2: Literature Review** – This section provides a detailed overview of learning style theories, the evolution of adaptive learning systems, and an in-depth review of semi-supervised learning methods previously applied in educational contexts.
- **Section 3: Dataset Description and Preprocessing** – Here, we describe the CSHS and CSMS datasets, detail how missing values were handled, explain the categorical encoding process, and introduce our new semi-supervised labeling model.
- **Section 4: Methodology** – This section presents the full processing pipeline, including hypothesis testing, synthetic sampling, feature engineering, selection, normalization, and the application of classification algorithms.
- **Section 5: Results and Analysis** – Once the experiments are complete, we will present our findings, comparative performance evaluations, statistical summaries, and visualizations.
- **Section 6: Conclusion and Future Work** – This final section will summarize our contributions, limitations, and offer directions for future enhancements such as real-time model deployment and application to broader academic domains.

## CHAPTER 2: Literature Review

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### 2.1 Survey of the State-of-the-art

**El Fazazi et al. (2021)** proposed a method to automatically detect a learner's learning style by analyzing web log files that reflect user behavior. These behavioral patterns were mapped to the Felder-Silverman Learning Style Model (FSLSM). The approach involved using a combination of the fuzzy C-means clustering algorithm and an artificial neural network to predict learning styles. Their model achieved a classification accuracy of 79.1%, indicating promising performance. However, the study highlighted some limitations, particularly in the reliability of web log mining. User behavior in logs can be ambiguous or misleading due to inconsistent or confusing interaction patterns. Despite this, the integration of AI techniques offers a significant step toward personalized e-learning environments [?].

**Alshammari and Qtaish (2019)** proposed an adaptive e-learning system that personalizes content based on a learner's knowledge level and learning style, specifically the information perception dimension from the Felder-Silverman model. The system customizes learning paths by sequencing learning objects according to whether students are sensory or intuitive learners and adjusts content delivery based on prior knowledge. A controlled experiment with 174 students demonstrated that learners who used the system adapted to both traits (learning style + knowledge level) achieved significantly higher learning gains than those adapted to only one trait. The highest post-test accuracy was observed in the combined group, with a mean score of 82.02%, compared to 72.16% (knowledge-level only) and 59.47% (learning-style only). The delayed post-test also confirmed sustained learning benefits. These findings highlight that dual-trait adaptation enhances both immediate and long-term learning accuracy and effectiveness.

**El-Sabagh et al. (2021)** proposed individual learning styles significantly influence how students absorb, process, and interpret information in the context of adaptive e-learning. Recognizing this, the study proposed a dynamic approach for identifying students' learning styles based on their interactions within a Moodle-based e-learning environment. Unlike traditional static questionnaires, this method relied on analyzing behavioral data—specifically, students' navigation patterns and activity logs during a Data Structures course. Learning styles were determined in alignment with the Felder-Silverman Learning Style Model (FSLSM) and then validated against quiz results administered at the course's end. The classification performance was evaluated using ROC (Receiver Operating Characteristic) curves, showing promising outcomes. The system achieved an average classification accuracy of 76%, indicating the effectiveness of behavior-based learning style prediction. The experiment involved 35 students, and multiple training-testing splits were applied, with accuracy varying slightly across FSLSM dimensions. This work underscores the potential of dynamic, data-driven techniques in personalizing e-learning experiences.

**Shamsuddin et al. (2010)** presented an intelligent method for identifying students' learning styles by analyzing interaction patterns within a Learning Management System (LMS). The approach leverages behavior-based data such as navigation paths, link selection, material preferences, and tool usage to map user characteristics into the Integrated Felder-Silverman (IFS) learning style model. The model expands the traditional FSLSM into sixteen distinct learning styles by combining four core dimensions. Using a rough set technique, twenty key behavioral attributes were selected to develop a rule-based classification framework. Since rough set analysis can produce numerous rules with potential redundancy, the study introduces a filtering process based on rule support, rule length, and classification accuracy. Through this filtering, the most significant and effective rules were isolated. The findings reveal that rules with lengths between 4 and 8 and support values ranging from 6 to 43 achieve the best results, yielding a high classification

accuracy of 96.62%. This demonstrates the effectiveness of combining rough set theory with learning analytics for precise learning style classification in e-learning environments.

**Traditional learning** style assessments, such as questionnaires, suffer from drawbacks like subjectivity and high time costs. Addressing these limitations, the study by **Wanying et al. (2024)** proposed an automated learning style detection method based on a Graph Attention Network (GAT), referred to as GAT-LS. The approach models student-learning material interactions as a bipartite graph, using node features derived from behavioral data to represent learners. GAT is employed to compute hidden vectors, capturing both global and local structural information. With a multi-head attention mechanism and a dropout-enhanced single-layer network, the model refines representations before applying K-means clustering to identify learning styles according to the Felder-Silverman Learning Style Model (FSLSM). Experiments conducted on the KDD CUP 2015 dataset yielded strong results, with the model achieving an accuracy of 96.47%, precision of 94.78%, recall of 91.71%, and F1 score of 93.46%, validating its effectiveness in detecting learning styles dynamically and at scale.

**Rasheed and Wahid (2021)** proposed an intelligent system to automatically detect learners' styles in e-learning environments by using behavioral data instead of traditional questionnaires. Their study integrated two popular learning style models—Gardner's theory of multiple intelligences and the Felder-Silverman Learning Style Model (FSLSM)—to develop a more comprehensive classification system. The authors extracted behavioral indicators from Moodle logs and implemented several machine learning algorithms, including Support Vector Machines (SVM), Random Forests, and Logistic Regression, to classify learners into their respective styles. The system was trained and tested on data from 498 students and showed promising results. For the FSLSM model, classification accuracy across the four dimensions ranged from 80% to 92%, with Decision Tree classifiers achieving the highest accuracy (92%) in the perception dimension and SVM performing best overall with consistent results across dimensions. The average consistency between the machine predictions and manual questionnaire-based labels was 86.75%, confirming the reliability of the proposed automated approach. These findings demonstrate the effectiveness of data-driven models in learning style detection, offering potential for more adaptive and personalized e-learning systems.

**Athanasios et al. (2025)** explored the application of Educational Data Mining (EDM) techniques to predict student performance and improve learning outcomes in higher education. Using data collected from 450 students across nine semesters in a Moodle-based Learning Management System (LMS), the study evaluated five machine learning algorithms: k-nearest neighbors (kNN), random forest, logistic regression, decision trees, and neural networks. The analysis revealed that identifying courses with strong correlations (above +0.3) significantly improved the accuracy of grade predictions, especially for binary classification tasks. Among the tested models, kNN and neural networks demonstrated superior performance, each achieving F1 scores above 0.8, highlighting their robustness and reliability. The results support the potential of EDM in enabling personalized learning and informing data-driven instructional strategies to foster student success.

**Bello and Wang (2024)** introduced an innovative method for detecting learners' learning styles by leveraging graph representation learning and machine learning techniques. Recognizing the importance of learning behavior analysis in online education, the authors propose a two-stage approach that first models student behavior as a bipartite graph, which is then transformed into low-dimensional embeddings via graph embedding techniques. These embeddings are used for downstream machine learning tasks. For learning style classification, the encoded learner sequences are clustered using the Fuzzy C-means algorithm, based on the Felder-Silverman Learning Style Model (FSLSM). Unlike earlier rigid, model-dependent approaches, this method offers flexibility and scalability in handling diverse online learner behaviors. Evaluated on the KDD Cup 2015 dataset, the proposed approach achieved a high average accuracy of 91.82% and precision of 87.37%, outperforming prior methods and proving its effectiveness in dynamically identifying learning styles to support adaptive learning environments.

**Yeime et al. (2023)** highlight the importance of identifying student learning styles for personalized education, especially within Adaptive Learning Systems (ALS). Previous studies have explored various classification techniques such as Decision Trees, Fuzzy C-Means, Artificial Neural Networks, Hidden Markov Models, and Genetic Algorithms for learning style prediction. Among these, Bayesian Networks (BNs) are emphasized for their strength in modeling uncertainty and incorporating expert knowledge. The approach demonstrated a high accuracy of approximately 90%.



Feldman et al. noted BNs as highly effective due to their probabilistic reasoning and capacity to handle incomplete data. Prior works using BNs mostly focused on a subset of FSLSM dimensions and used limited data samples, often lacking comprehensive validation. Studies like Azzi et al. and Abdullah et al. contributed by applying FCM and DTs in LMS environments. However, comparative evaluations remain challenging due to variability in datasets and evaluation metrics. The reviewed works collectively identify a gap in the scalability and precision of learning style classification, which this paper aims to address using Object-Oriented Bayesian Networks (OBNs) with both full and shortened ILS questionnaires.

**Sheeba and Reshmy (2018)** address the significance of learning styles in enhancing personalized learning within Learning Management Systems (LMS). It reviews existing automatic learning style detection methods, emphasizing decision tree classifiers as widely accepted tools due to their interpretability and effectiveness. However, earlier models often suffer from weak decision rules, leading to lower classification accuracy. To overcome this, the paper proposes a refined decision tree approach using strong, behavior-based rules extracted from LMS log files. The classification is grounded in the Felder-Silverman Learning Style Model (FSLSM), focusing on three key dimensions: processing, perception, and input. Previous works have relied on static data, whereas this approach dynamically analyzes real student interactions. The system employs forward reasoning inference to match behavior patterns with learning styles. The experimental results, tested on data from 300 students in Moodle, demonstrate improved prediction accuracy, achieving an average of 87%, especially stable in perception and input dimensions.

**Samina and Noor (2016)** explored the automatic detection of learning styles in Learning Management Systems (LMS) to enhance personalized e-learning. Traditional learning style identification, such as Kolb's Learning Style Inventory (KLSI), relies on manual questionnaires, which can be time-consuming and intrusive. To address this, the study proposes a Bayesian Network (BN)-based data mining approach to infer learning styles directly from students' interaction data on LMS platforms. A large dataset was used to map classroom learning preferences into the e-learning context. The BN model outputs probabilistic values that act as thresholds for classifying learning styles. An experimental comparison with KLSI in a Pakistani university showed that the automatic method achieved an accuracy of approximately 71%. This method offers a promising, scalable alternative for adaptive learning environments. The authors suggest further improvements by comparing BN with other classification algorithms to enhance precision.

**Reshmy et al. (2019)** focused on the automatic detection of students' learning styles in Learning Management Systems (LMS) using a Decision Tree classification algorithm. The approach aims to enhance adaptive learning by analyzing student interaction data and predicting their learning styles without the need for manual questionnaires. The classification is based on selected behavioral indicators aligned with the Felder-Silverman Learning Style Model (FSLSM). The model was evaluated with data from LMS usage and achieved a high overall accuracy of 87%. However, the study only considered three of the four FSLSM dimensions, omitting two key learning style components. Additionally, the results were not benchmarked against other classification techniques. Despite these limitations, the findings support the feasibility of using decision trees for real-time learning style prediction in e-learning environments.

**Azzi and Jeghal (2019)** proposed a robust classification approach for predicting learning styles in adaptive e-learning systems using the Fuzzy C-Means (FCM) clustering algorithm. The study focuses on enhancing the adaptability of learning environments by clustering learners based on interaction behaviors and learning preferences. The FCM method was applied to categorize students according to the Felder-Silverman Learning Style Model (FSLSM), aiming to personalize content delivery. The approach demonstrated a high accuracy of approximately 93%, highlighting its effectiveness in recognizing distinct learning patterns. However, the study did not compare the results against other classification techniques, leaving room for further evaluation of its relative performance. The results underscore the potential of FCM for use in scalable and intelligent adaptive e-learning systems.

**Sarimehmet et al. (2023)** investigated the dominant learning styles of nursing students and the factors influencing them using decision tree algorithms. The study involved 421 nursing students from a university in Turkey and utilized CART, CHAID, and QUEST decision tree models. Results showed that visual (48%) and tactile (36%) were the most common learning styles. The analysis revealed that living place and university entrance score had statistically significant effects on learning style classification ( $p < 0.05$ ), with living place being the most influential variable.

The study did not report classification accuracy or compare with other machine learning methods, but effectively demonstrated the interpretability and usefulness of decision trees in educational profiling. The findings support the need for learning environments tailored to the varied styles of nursing students to enhance clinical education outcomes.

**Karagiannis and Satratzemi (2018)** proposed an automatic learning style detection approach to enhance adaptive learning in Moodle-based environments. Their method combines data from the Index of Learning Styles (ILS) questionnaire with learners' behavioral patterns and interactions within Moodle. The classification accuracies achieved for the four dimensions of the Felder-Silverman Learning Style Model (FSLSM) were 70%, 66%, 75%, and 80%, respectively. While the approach demonstrated reasonable effectiveness, it faces several limitations. These include the inherent inaccuracy of self-reported questionnaire data, dependency on Moodle logs, high computational complexity, and limited scalability. Additionally, the system lacks flexibility, involves time-intensive adaptation processes, and struggles with detecting imbalanced learning styles effectively.

**Barros De Castro** provides insights into identifying students' learning styles in distance education by analyzing their interactions with a Learning Management System (LMS). The study looks at behavioral patterns such as time spent on resources, forum involvement, and assessment activities using data mining techniques like decision trees and clustering algorithms. To properly categorize students, these patterns are matched to recognized learning style models. The study emphasizes how LMS data can provide useful markers of unique learning preferences, allowing for more individualized instruction. Notwithstanding encouraging outcomes, the study admits its shortcomings, such as uneven student participation and the requirement for more data sources to increase precision. It highlights how intelligent systems might facilitate adaptive learning and suggests more study to improve model scalability and reliability across various learning environments.

**Ali and Iqbal (2024)** introduced a hybrid deep learning framework that integrates Learning Style Identification and Sentiment Analysis to enhance adaptability in smart education systems. The model combines CNN-BiLSTM and Transformer architectures to classify learning styles and extract sentiments from learners' feedback. Their method achieved an accuracy of 92.4%, showcasing the model's strong performance in personalized content delivery. A key limitation was the computational complexity and lack of real-time deployment testing. Nevertheless, the approach holds promise for intelligent, learner-centered e-learning platforms.

**Graf et al. (2009)** proposed an automatic student modeling approach within Learning Management Systems (LMS) to identify learners' styles based on the Felder-Silverman Learning Style Model. The system used behavioral tracking data to infer styles without requiring explicit learner input. Accuracy of the model reached approximately 84%, indicating solid predictive capability. However, the authors acknowledged limitations in generalizability across diverse learning contexts and the need for further validation with larger, heterogeneous datasets.

**Shih, Feng, and Tsai (2008)** examined adaptive e-learning systems that personalize content based on learners' styles using a rule-based method aligned with the Felder-Silverman Learning Style Model. The system achieved an accuracy of 87%, indicating strong performance in adapting to individual preferences. However, reliance on fixed rules limited its scalability and responsiveness to dynamic learner behavior. The study also noted issues with inaccurate self-assessment affecting system performance. Despite these constraints, the research offers valuable direction for future systems, recommending the use of machine learning for more flexible personalization.

**Roseli et al.** emphasize the value of recognizing individual learning styles to enhance the effectiveness of medical education. They highlight the shift from passive to active, student-centered learning and stress the need for educators to understand and adapt to students' diverse learning preferences. By identifying whether learners have unimodal or multimodal styles, teachers can develop tailored strategies that improve knowledge retention and application. The authors also point out that in Brazil, reflective teaching practices in higher education are still evolving. They argue that identifying learning styles is a beneficial and practical approach to improving instructional quality, fostering a more inclusive academic environment, and ultimately enhancing the performance and engagement of medical students.

**Fareeha et al.** review the evolution of automatic detection of learning styles, addressing limitations of traditional questionnaires such as low student motivation and self-awareness. They analyze various computational approaches—like machine learning, data mining, and user modeling—that infer learning preferences from user interactions in digital environments. The study highlights the potential of these methods to personalize learning experiences, thereby enhancing educational outcomes. However, the authors also note challenges, including data privacy concerns and the need for more robust validation across diverse educational settings. They conclude by emphasizing the importance of interdisciplinary research to refine these technologies for broader application in adaptive learning systems.

**Christos Troussas et al.** introduce an ensemble learning approach utilizing fuzzy weights to enhance the identification of students' learning styles in adaptive instructional systems. By combining multiple classifiers, the model achieved a high accuracy rate of 94.44%, surpassing previous methods that reported accuracies of 85% and 80%. This significant improvement underscores the model's effectiveness in accurately detecting diverse learning preferences. The study emphasizes the importance of precise learning style identification in developing personalized educational experiences, which can lead to increased student engagement and improved academic outcomes. The authors advocate for the integration of such advanced ensemble learning techniques into e-learning platforms to foster more tailored and effective educational environments.

**Hanaa El Fazazi et al.** propose a novel method for identifying students' learning styles within adaptive e-learning systems by analyzing web log data. This approach maps learners' behaviors to the Felder-Silverman Learning Style Model using the Fuzzy C-Means clustering algorithm, effectively categorizing students based on their interaction patterns. To accommodate the dynamic nature of learning preferences, the system incorporates an Artificial Neural Network that predicts and adapts to changes in learning styles over time. This dual-method framework enhances the personalization of educational content, aiming to improve learner engagement and academic performance. The study underscores the importance of integrating adaptive mechanisms in e-learning platforms to cater to individual learning needs effectively.

**Yu Hsin Hung et al.** look into how comprehension results are affected when teaching strategies are matched to students' self-reported learning preferences. They discovered that matching teaching strategies to students' chosen learning styles did not significantly improve comprehension, indicating that this alignment does not increase the efficacy of learning. The study promotes evidence-based teaching strategies that accommodate a range of learning demands and highlights that depending solely on learning style evaluations may not be a trustworthy way to enhance educational achievements.

**Glazunova et al.** explored whether matching teaching methods to students' preferred learning styles—such as visual, auditory, or kinesthetic—improves their understanding of the material. The researchers divided students based on their self-identified learning styles and then taught them using methods that aligned or did not align with these preferences. Despite these adjustments, the study found no significant improvement in students' comprehension, regardless of whether the teaching methods matched their preferred learning styles. This suggests that tailoring instruction to individual learning styles may not be an effective strategy for enhancing learning outcomes. The study advocates for using evidence-based teaching methods that address the diverse needs of all students.

**Jason Bernard et al.** used the VARK model and data analysis techniques demonstrated a high level of accuracy in classifying learning preferences. Specifically, the system achieved reliability coefficients ranging from 0.77 to 0.85 for the VARK sub-scales, indicating strong internal consistency. Additionally, a semi-supervised machine learning approach applied to the Felder-Silverman learning style model achieved classification accuracies of 88.83% and 77.35% across two courses, respectively. These findings suggest that the system is effective in accurately identifying students' learning styles. However, it's important to note that while the system's accuracy is high, the practical benefits of aligning instructional methods with students' self-identified learning styles in terms of improved learning outcomes remain inconclusive. Research indicates that matching teaching methods to learning styles does not consistently enhance educational outcomes.

### 2.1.1 Summary

A significant body of research has explored the automatic detection of learning styles to improve personalized learning in digital environments. Early approaches, such as those by El Fazazi et al. [?] and Shamsuddin et al. [?], leveraged behavioral data like web logs and navigation paths to classify learners based on the Felder-Silverman Learning Style Model (FSLSM). Techniques including fuzzy C-means, artificial neural networks, and rough set theory were employed, achieving accuracies ranging from 79.1% to 96.62%. These studies highlighted the advantages of data-driven learning style detection over traditional questionnaires, although issues like noisy data and rule redundancy were noted. Alshammari and Qtaish [?] further emphasized the importance of combining learning style adaptation with knowledge-level personalization to improve learning gains.

Recent advancements introduced sophisticated models using graph-based learning and ensemble techniques. Wanying et al. [?] proposed a GAT-based model that represented learner-material interactions as bipartite graphs and achieved over 96% accuracy. Similarly, Bello and Wang [?] employed graph embedding combined with fuzzy clustering for flexible, scalable classification. Other studies, such as those by Rasheed and Wahid [?], used hybrid models integrating FSLSM with Gardner’s theory and achieved high predictive reliability using machine learning techniques like SVM and decision trees. Despite strong classification accuracy, some studies pointed out that dynamic behavioral tracking could outperform static questionnaires only under certain conditions, emphasizing the need for robust data handling and model interpretability.

Several researchers explored educational data mining (EDM) to enhance learning outcomes. Athanasios et al. [?] demonstrated the utility of machine learning algorithms such as kNN and neural networks in predicting student performance, supporting the broader role of learning analytics in personalization. Others, like Sheeba and Reshmy [?], focused on refining decision tree rules based on LMS data, enhancing accuracy while maintaining model transparency. Still, studies by Azzi and Jeghal [?], and Samina and Noor [?] highlighted the potential of clustering and Bayesian networks in LMS environments, though they called for stronger comparative evaluations and broader FSLSM coverage.

While many approaches showed strong predictive performance, some studies raised doubts about the educational impact of matching instruction to learning styles. Yu Hsin Hung et al. [?] and Glazunova et al. [?] found that aligning teaching methods with self-reported learning styles did not significantly enhance comprehension. These findings caution against relying solely on learning style classification and advocate for evidence-based, inclusive teaching methods. Overall, the reviewed literature reflects a shift from static assessments to intelligent, data-driven models, while also acknowledging the ongoing debate over the practical benefits of learning style personalization in education.

Table 2.1: List of the Related Works

| Study                        | Approach                                     | FSLSM           | Techniques Used                       | Performance                                             | Data Source               | Key Limitations                                               |
|------------------------------|----------------------------------------------|-----------------|---------------------------------------|---------------------------------------------------------|---------------------------|---------------------------------------------------------------|
| El Fazazi et al. (2021)      | Web log-based learning style detection       | Yes             | Fuzzy C-Means, ANN                    | 79.1% Accuracy                                          | Web logs                  | Ambiguity in user behavior; noisy data                        |
| Alshammari and Qtaish (2019) | Adaptive e-learning system using dual traits | Yes             | Rule-based sequencing, questionnaires | 82.02% (dual-trait), 72.16% (knowledge), 59.47% (style) | LMS (174 students)        | Limited to one FSLSM dimension; rule-based logic not scalable |
| El-Sabagh et al. (2021)      | Behavior-based dynamic style detection       | Yes             | Activity log analysis, ROC evaluation | ~76% Accuracy                                           | Moodle logs (35 students) | Small sample size; limited generalization                     |
| Shamsuddin et al. (2010)     | Intelligent learning style detection via LMS | Yes (IFS model) | Rough Set Theory, Rule Filtering      | 96.62% Accuracy                                         | LMS logs                  | Rule redundancy; required filtering                           |

| Study                             | Approach                                  | FSLSM             | Techniques Used                        | Performance                       | Data Source                   | Key Limitations                                   |
|-----------------------------------|-------------------------------------------|-------------------|----------------------------------------|-----------------------------------|-------------------------------|---------------------------------------------------|
| Wanying et al. (2024)             | Automated detection using graph models    | Yes               | Graph Attention Network (GAT), K-Means | 96.47% Accuracy, 94.78% Precision | KDD CUP 2015                  | Needs structured graph data                       |
| Rasheed and Wahid (2021)          | ML-based style detection using LMS logs   | Yes + Gardner     | SVM, RF, Logistic Regression           | 80–92% across FSLSM dimensions    | LMS logs (498 students)       | Requires hybrid model; depends on consistent logs |
| Athanasios et al. (2025)          | EDM for predicting performance            | No                | kNN, RF, DT, NN, Logistic Regression   | F1 > 0.8 (kNN, NN)                | Moodle logs (450 students)    | Focuses on grades, not styles                     |
| Bello and Wang (2024)             | Graph-based style detection with ML       | Yes               | Graph Embedding + FCM Clustering       | 91.82% Accuracy, 87.37% Precision | KDD CUP 2015                  | High complexity; limited interpretability         |
| Yeime et al. (2023)               | Probabilistic modeling for ALS            | Yes               | Bayesian Networks                      | ~90% Accuracy                     | LMS logs + ILS questionnaires | Limited FSLSM coverage; inconsistent validation   |
| Sheeba and Reshmy (2018)          | Refined DT-based learning style detection | Yes               | Decision Trees, forward inference      | 87% Accuracy                      | Moodle logs (~300 students)   | Focuses on 3 FSLSM dimensions only                |
| Samina and Noor (2016)            | BN-based automatic detection              | No (Kolb's model) | Bayesian Networks                      | ~71% Accuracy                     | LMS data (Pakistan)           | Not benchmarked against other classifiers         |
| Reshmy et al. (2019)              | DT-based style detection                  | Yes               | Decision Trees                         | 87% Accuracy                      | LMS logs                      | Only 3 FSLSM dimensions considered                |
| Azzi and Jeghal (2019)            | Clustering learners in e-learning         | Yes               | Fuzzy C-Means                          | ~93% Accuracy                     | LMS behavior logs             | Not compared with other techniques                |
| Sarimehmet et al. (2023)          | Style profiling for nursing students      | No                | CART, CHAID, QUEST                     | N/A (Descriptive focus)           | Survey (421 nursing students) | No accuracy reported; non-ML validation           |
| Karagiannis and Satratzemi (2018) | Style detection using Moodle logs + ILS   | Yes               | ILS Questionnaire + LMS Behavior       | 70%, 66%, 75%, 80%                | Moodle data                   | High computational cost, limited scalability      |
| Barros De Castro (2016)           | LMS interaction-based style detection     | Yes               | DT, Clustering                         | Not specified                     | LMS logs                      | Limited sources; uneven participation             |
| Ali and Iqbal (2024)              | DL for style + sentiment                  | Yes               | CNN-BiLSTM, Transformer                | 92.4% Accuracy                    | Feedback + behavioral data    | High computational cost; not tested live          |

| Study                          | Approach                               | FSLSM | Techniques Used             | Performance                     | Data Source             | Key Limitations                           |
|--------------------------------|----------------------------------------|-------|-----------------------------|---------------------------------|-------------------------|-------------------------------------------|
| Graf et al. (2009)             | Behavior-based modeling in LMS         | Yes   | Behavior Tracking           | ~84% Accuracy                   | LMS                     | Generalization across contexts limited    |
| Shih et al. (2008)             | Rule-based adaptive learning           | Yes   | Fixed Rules                 | 87% Accuracy                    | FSLSM questionnaire     | Lacks dynamic adaptability                |
| Roseli et al. (N/A)            | Review in medical education            | Yes   | Literature Review           | N/A                             | Medical student context | Theoretical focus; no evaluation          |
| Fareeha et al. (N/A)           | Review of automated detection          | Yes   | ML, Data Mining             | Not specified                   | Literature              | Lacks empirical testing; privacy concerns |
| Tzu-Chien-Leu (2008)           | Evaluation of ILS in e-learning        | Yes   | ILS                         | 75% Accuracy                    | E-learning platforms    | No adaptive mechanism; early-stage        |
| Christos Troussas et al. (N/A) | Ensemble learning with fuzzy weighting | Yes   | Ensemble ML, Fuzzy Weights  | 94.44% Accuracy                 | Not specified           | High model complexity                     |
| Hanaa El Fazazi et al. (N/A)   | Dual-method AI framework               | Yes   | FCM + ANN                   | Not specified                   | Web logs                | No specific results given                 |
| Yu Hsin Hung et al. (N/A)      | Matching strategies study              | Yes   | Teaching Strategy Alignment | No improvement in comprehension | Questionnaire based     | Style-matching ineffective                |
| Glazunova et al. (2020)        | Teaching style alignment analysis      | Yes   | Manual Matching             | No significant gain             | Self-identified styles  | Matching styles not useful                |
| Jason Bernard et al. (2017)    | Semi-supervised detection              | Yes   | Semi-supervised ML          | 88.83%, 77.35% (2 courses)      | VARK + FSLSM            | Unclear impact on outcomes                |

## CHAPTER 3: Materials and Method

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### 3.1 Materials

The research adopts an analysis and classifies students' learning styles based on their LMS behavior using the FSLSM model. Two datasets were collected from a Moodle-based LMS platform, each representing students from different academic backgrounds—medical and humanities. These datasets consist of behavioral interaction data, including activity logs, quiz performance, and time spent on course materials. The CSMS dataset includes 670 samples (256 labeled, 414 unlabeled), while the larger CSHS dataset contains 1952 samples (727 labeled, 1225 unlabeled). Each dataset consists of 17 features describing student interactions with course materials. The datasets were carefully prepared to ensure quality and consistency before model development, allowing for a comparative analysis of learning patterns across disciplines. All experiments and analyses were conducted in a cloud-based environment using Google Colab, leveraging Python libraries such as pandas, NumPy, and scikit-learn to ensure efficient data handling and reproducibility.

#### 3.1.1 Dataset Collection

The datasets used in this study were collected from two Moodle-based university courses: Computer Skills for Medical Students (CSMS) and Computer Skills for Humanities Students (CSHS). These courses were chosen due to their structured learning environments and the availability of comprehensive log data. Each dataset includes both labeled and unlabeled samples. The labeled data contains students' learning style categories based on the Felder-Silverman Learning Style Model (FSLSM), obtained through responses to the Index of Learning Styles (ILS) questionnaire. The unlabeled data consists of the same behavioral features without corresponding questionnaire responses. Student activity logs were automatically captured by the Moodle Learning Management System (LMS) while students interacted with various course materials, such as reading files, video lectures, self-assessments, and quizzes. These logs serve as behavioral indicators that reflect different learning preferences. The collected data was then cleaned, organized, and formatted into structured datasets for further analysis. This method of combining self-reported learning styles with system-generated behavior data provides a rich foundation for developing models that detect learning styles without relying solely on questionnaires.

#### 3.1.2 Dataset Exploration

After collecting the datasets, an initial exploration was conducted to understand the structure, quality, and distribution of the data. Both the CSMS and CSHS datasets consist of 17 features, including 12 numerical and 5 categorical variables that reflect various student activities within the LMS—such as reading course materials, watching videos, taking quizzes, and completing exercises. The datasets also include four target variables corresponding to the FSLSM dimensions: Processing, Perception, Input, and Understanding.

Exploratory analysis revealed key differences between the two datasets. The CSHS dataset contains a total of 1952 samples (727 labeled and 1225 unlabeled), while the CSMS dataset includes 670 samples (256 labeled and 414 unlabeled), making CSHS nearly three times larger. We also observed class imbalance in certain learning style categories, particularly in the Perception variable of the CSHS dataset. In contrast, the CSMS dataset has more balanced distributions for most learning style labels but lacks variation in the Understanding dimension, which includes only one class (GLO).

Additionally, both datasets showed no missing values in the labeled samples, and the ‘id’ feature was removed as it only contained unique identifiers with no analytical value. This exploration stage helped identify potential challenges such as class imbalance and uninformative features, which guided our preprocessing and modeling steps.

### 3.1.3 Dataset Sampling

Both datasets consist of labeled and unlabeled samples, making them suitable for semi-supervised learning. In the CSMS dataset, 256 out of 670 samples are labeled, while the remaining 414 are unlabeled. Similarly, in the CSHS dataset, 727 samples are labeled and 1225 are unlabeled. To ensure meaningful training and evaluation, only the labeled data were initially used to build baseline models. The unlabeled data were later incorporated using semi-supervised techniques to enhance model generalization and performance, especially in the presence of class imbalance and limited labeled instances.

### 3.1.4 Dataset Processing

Before training the models, both the CSMS and CSHS datasets underwent essential preprocessing steps to ensure data quality and model readiness. First, the “id” feature was removed from both datasets because it contained unique identifiers, which did not contribute to the learning style prediction task. In the CSMS dataset, the “Understanding” feature was also dropped, as it contained only a single category (GLO), rendering it non-informative for classification purposes.

Next, all categorical features were transformed into a numerical format using suitable encoding techniques to make them compatible with machine learning algorithms. Additionally, all numerical features were normalized to bring them onto a similar scale, which helps improve model convergence and performance, especially for distance-based and gradient-based methods.

As part of data cleaning, we also removed any rows that contained four or more zero values, assuming that such entries indicated incomplete or inactive student participation and could negatively impact model training. After cleaning, the labeled data was used for training and validation in supervised learning tasks, while the unlabeled data was leveraged in semi-supervised learning to enhance prediction accuracy by utilizing the full dataset more effectively.

### 3.1.5 Research Environment and Devices

All experiments were conducted using Google Colab, a cloud-based platform providing free access to GPU resources for faster computation. The programming was done in Python, leveraging libraries for data handling, machine learning, and visualization. Jupyter notebooks provided an interactive environment for model development and evaluation. The hardware used included a standard Google Colab runtime, ensuring efficient execution of both traditional machine learning and deep learning models.

#### Hardware:

- **GPU:** NVIDIA Tesla T4 GPU
- **RAM:** 16 GB
- **Processor:** AMD Ryzen 5

#### Software:

- **Platforms:** Google Colab
- **Libraries:** pandas, NumPy, scikit-learn, matplotlib

#### Operating System:

- **Operating System:** Windows 11

This environment provided a practical and accessible solution for performing complex machine learning tasks, enabling us to develop, test, and refine our models effectively.



## 3.2 Methods

Here we discussed our proposed model for this project. We mentioned the framework and algorithm. We showed the experimental setup for this project.

### 3.2.1 Proposed Models

In our study, we focused on the automatic detection of learning styles within e-learning environments by leveraging machine learning models trained on student interaction data. Our objective was to create a model that could accurately classify learners based on the Felder-Silverman Learning Style Model (FSLSM) using behavioral data. To achieve this, we followed a comprehensive pipeline of preprocessing, model experimentation, and evaluation.

We began by cleaning the dataset through missing value imputation, encoding categorical features, and standardizing numerical values. A key challenge encountered was the class imbalance across learning style categories. To address this, we applied the Synthetic Minority Oversampling Technique (SMOTE), which improved model learning by generating synthetic samples for underrepresented classes and reducing classification bias.

We experimented with several machine learning algorithms, including:

- XGBoost
- Random Forest
- Decision Tree
- K-Nearest Neighbors (KNN)
- Logistic Regression

Among these, XGBoost and Random Forest consistently achieved the highest performance across all FSLSM dimensions. XGBoost stood out due to its gradient boosting framework and effective regularization, while Random Forest delivered strong accuracy through ensemble learning and robust handling of non-linear relationships. Both models showed resilience to overfitting and performed well with imbalanced data.

Performance evaluation was based on standard metrics such as Accuracy, Precision, Recall, F1-score, and ROC-AUC. We also applied 10-fold cross-validation to ensure model robustness and minimize sampling bias. Confusion matrices were analyzed to gain insight into misclassifications, and the use of hyperparameter tuning (via GridSearchCV and RandomizedSearchCV) helped optimize each model's performance.

Models like KNN and Logistic Regression underperformed in our tests, particularly in high-dimensional or imbalanced scenarios, due to their sensitivity to data distribution and lack of complexity to capture nuanced patterns.

In conclusion, our proposed model integrates SMOTE for data balancing and leverages XGBoost and Random Forest as the primary classifiers. This combination delivers high accuracy and interpretability, making it well-suited for real-world deployment in adaptive and personalized e-learning platforms. The system eliminates the need for manual surveys and offers a scalable solution for understanding and supporting diverse learner needs.

### 3.2.2 Design/Framework

We utilized a dataset containing a range of features related to learners' engagement and behavior in an e-learning environment. These features included forum interactions, quiz performance, content viewing patterns, and demographic attributes such as gender. The target outputs were learners' styles across four FSLSM (Felder-Silverman Learning Style Model) dimensions: active-reflective, sensing-intuitive, visual-verbal, and sequential-global. These labels were derived from self-reported ILS questionnaires, and since each dimension is binary, the overall task was framed as a set of independent binary classification problems.

The entire implementation was carried out using Google Colab to ensure seamless execution and scalability. Essential Python libraries were imported, including pandas, numpy, matplotlib, tensorflow, keras, and scikit-learn, along with model-specific packages such as xgboost. Data preprocessing was a key step in the pipeline. This involved handling missing values, encoding categorical variables, and normalizing numerical features to a standard scale. These steps were necessary to ensure compatibility with various machine learning algorithms and to enhance convergence behavior during training.

A major challenge encountered was class imbalance, particularly in the learning style labels. To address this, the Synthetic Minority Oversampling Technique (SMOTE) was applied exclusively to the training data. SMOTE artificially generates new instances of minority classes by interpolating between existing samples, thus helping the models learn decision boundaries more effectively without bias toward the majority class.

Once preprocessing was complete, we developed and trained a range of classification models. Each model was trained on the balanced dataset, and performance was evaluated on a held-out test set using accuracy, F1-score, precision, recall, and ROC-AUC as evaluation metrics. 10-fold cross-validation was used across all experiments to ensure robust generalization and to reduce dependency on any particular data split.

Among the algorithms tested, XGBoost consistently delivered superior performance, attributed to its boosting-based structure, regularization, and inherent handling of class imbalance. Random Forest and Decision Tree models followed closely with respectable accuracy and good interpretability. However, Logistic Regression and KNN underperformed in comparison, likely due to their sensitivity to noise and limitations in capturing non-linear patterns in high-dimensional behavioral data.

Although we did not integrate SHAP (SHapley Additive exPlanations) in this phase, it remains a promising tool for future implementation to enhance model transparency. Such interpretability would allow us to pinpoint which learner behaviors most strongly influence the predicted learning styles, thus making the framework more explainable and applicable in real educational settings.

In conclusion, the proposed design integrates systematic data preprocessing, effective resampling techniques, and comparative model training to construct a robust classification pipeline. It provides a foundational architecture that can be extended and embedded into adaptive e-learning systems, enabling dynamic content personalization based on students' individual cognitive preferences.

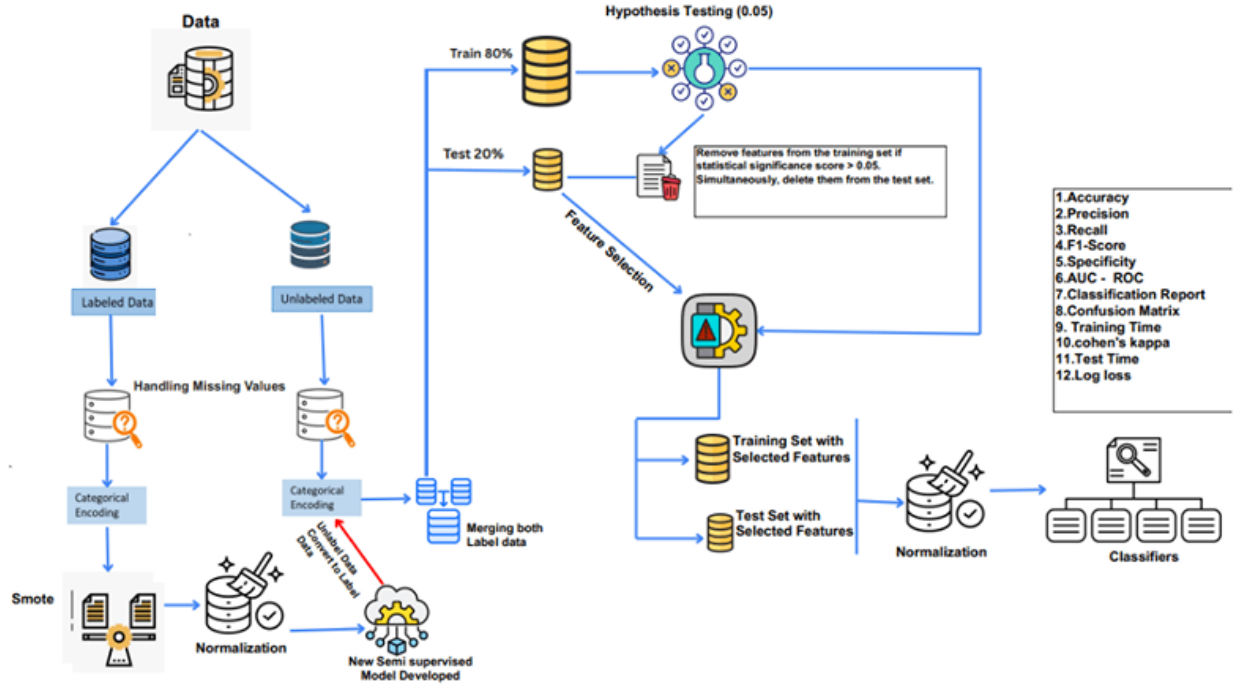


Figure 3.1: Overview of the Proposed Method

### 3.2.3 Algorithms/Models Formulation

In the framework of our learning style prediction system, we employed a range of machine learning models, each grounded in distinct theoretical formulations and computational mechanisms. Our intention was to rigorously evaluate the effectiveness of these algorithms in handling behavioral data extracted from an e-learning environment, especially considering challenges like non-linearity, class imbalance, and feature interdependence.

We first considered Logistic Regression, a generalized linear model used for binary classification. It models the log-odds of the probability that a student belongs to a particular learning style class using a linear combination of input features. The model uses the sigmoid function:

**fixed** This output is interpreted as the probability that the instance belongs to the positive class. Logistic regression is highly interpretable, but its linear decision boundary limits its performance in data distributions that require non-linear separation. This output is interpreted as the probability that the instance belongs to the positive class. Logistic regression is highly interpretable, but its linear decision

This output is interpreted as the probability that the instance belongs to the positive class. Logistic regression is highly interpretable, but its linear decision boundary limits its performance in data distributions that require non-linear separation.

We then explored Naive Bayes, a probabilistic classifier based on Bayes' Theorem, which assumes that all features are conditionally independent given the class label:

**fixed** This classifier works particularly well with high-dimensional datasets and is computationally efficient, but its strong independence assumption often fails in real-world educational data where behavioral features (like quiz attempts and content views) are interdependent. As a result, it underperformed in our experiments. This classifier works particularly well with high-dimensional datasets and is ca where behavioral features (like quiz attempts and content views) are interdependent. As a result, it underperformed in our experiments.

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The K-Nearest Neighbors (KNN) algorithm provides a non-parametric method of classification that does not learn an explicit model but instead memorizes the training dataset. Classification is done by identifying the k closest data points (based on Euclidean or other distance metrics) and taking a majority vote:

**FixedKNN** is simple and intuitive but becomes computationally expensive with large datasets and is sensitive to irrelevant features and feature scaling. Moreover, after applying SMOTE, the presence of synthetic data further affected the local neighborhood structure, making KNN less reliable. KNN is simple and intuitive but becomes computationally expensive with large datasets and is sensitive to irrelevant features and feature scaling. Moreover, after applying SMOTE, the p data further affected the local neighborhood structure, making KNN less reliable.

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To capture hierarchical and non-linear patterns in the data, we implemented Decision Trees, which split the data into subsets based on feature values that maximize information gain (e.g., Gini Index or Entropy). A typical decision rule is formed by recursive partitioning:

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Decision Trees provide a clear interpretation of how decisions are made, and they performed well in our scenario. However, they tend to overfit, especially when the tree grows deep without pruning.

Random Forest benefited from reduced variance and performed consistently well across different learning style labels, offering a strong balance between accuracy and interpretability.

The standout model in our research was XGBoost (Extreme Gradient Boosting), which iteratively builds decision trees by optimizing a regularized objective function that balances predictive accuracy and model complexity. The overall prediction is computed by summing the outputs of all weak learners:

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XGBoost introduces regularization parameters ( $\gamma$ ,  $\lambda$ ) that penalize model complexity, making it particularly robust to overfitting. It handles missing data internally, supports parallel processing, and integrates tree pruning. These properties allowed XGBoost to outperform other models in accuracy, F1-score, and overall robustness, especially after applying SMOTE to deal with class imbalance.

We also experimented with Support Vector Machines (SVM), which aim to find the hyperplane that maximally separates classes in the feature space. For linearly separable data, the optimization is:

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For non-linearly separable data, SVM employs kernel functions to transform data into higher dimensions, where a linear separator can be applied. While SVM showed moderate success, its performance degraded on large and imbalanced datasets, and training times became lengthy due to kernel computation and the synthetic data introduced by SMOTE.

Through comparative evaluation using cross-validation, confusion matrices, and classification metrics (accuracy, precision, recall, F1-score, ROC-AUC), we identified XGBoost and Random Forest as the top-performing models. They effectively captured complex behavioral patterns in the data and generalized well across all learning style dimensions. Meanwhile, simpler models like Logistic Regression and Naive Bayes struggled with the intricate, imbalanced nature of the data, and KNN and SVM exhibited sensitivity to noise and scale.

In conclusion, this systematic evaluation of models revealed that tree-based ensemble methods, particularly XGBoost, form a robust foundation for predicting learner profiles in adaptive educational systems. Their mathematical flexibility and regularization mechanisms make them ideal for complex, imbalanced, and high-dimensional data commonly found in e-learning platforms.

### 3.2.4 Experimental Setup

To implement and test the classification of learning styles, the experiments were performed in a Python environment using Jupyter Notebook. The dataset, which contained students' behavioral and performance data, was loaded and processed using pandas for structured handling. Prior to modeling, all features were assessed for consistency, and any missing values were addressed through simple imputation. Categorical variables were transformed using label encoding, and numerical variables were normalized to ensure compatibility with algorithms sensitive to feature scaling.

The dataset showed clear signs of class imbalance, particularly in the distribution of learning styles, which could have biased model training. To overcome this, we incorporated SMOTE, which synthetically generates new samples for the minority classes by interpolating between existing points in the feature space. This method was applied strictly to the training data to prevent data leakage into the evaluation phase.

A train-test split of 80:20 was used to separate training and evaluation sets. We developed and tested a diverse set of machine learning models including logistic regression, naive Bayes, K-nearest neighbors (KNN), decision tree, random forest, support vector machine (SVM), and XGBoost. Each algorithm was implemented using scikit-learn or XGBoost libraries and configured with a baseline set of hyperparameters. Further tuning was carried out using cross-validation combined with randomized or grid search, depending on the complexity of the model.

Cross-validation was consistently applied (10-fold) to ensure stable and generalized performance estimation across models. Each model's outputs were evaluated using classification metrics such as accuracy, precision, recall, F1-score, and ROC-AUC. Confusion matrices were also visualized to understand class-level prediction behavior. All modeling steps, from preprocessing to evaluation, were executed in a controlled notebook environment to ensure reproducibility and traceability of results across experimental runs.

## 3.3 Implementation

To demonstrate the practical application of our learning style classification model, we developed a user-friendly website that allows users to interact directly with the prediction system. The primary functionality begins with the selection of a target learning style dimension. Users can choose one of the four main categories: Processing, Perception, Input, or Understanding. Once selected, the system prepares to classify that particular target variable based on the user-provided input features.

After selecting the target, users are prompted to enter values for the remaining features—such as course activity metrics, video interaction behavior, and engagement with course materials. These inputs are passed to the backend, where our trained model processes the data and predicts the category (e.g., ACT or REF for Processing). This real-time prediction offers immediate feedback on the student's likely learning style.

To enhance interpretability, the website displays prediction results using bar charts and line graphs. These visualizations help users understand the distribution and confidence of the model's outputs. Users can experiment with different values to observe how predictions change dynamically, making the tool both educational and exploratory.

Furthermore, the application includes a feature importance display, which highlights which inputs most influenced the prediction. This adds transparency to the model's decision-making and offers users a clearer understanding of the relationship between LMS behaviors and learning styles.

The backend of the website is built using Flask, which connects the frontend interface with the machine learning model. This enables seamless data flow and live processing, ensuring smooth user interactions and real-time prediction results. The combination of interactivity, visualization, and interpretability makes this website a practical extension of our research and a valuable tool for students, educators, and researchers alike.

## 3.4 Summary

This research presented the complete workflow for developing a machine learning-based system to detect students' learning styles from LMS behavioral data. The materials included two datasets: Computer Skills for Humanities Students (CSHS) and Computer Skills for Medical Students (CSMS), collected from Moodle-based courses. These datasets captured detailed user interactions and were pre-labeled using responses based on the Felder-Silverman Learning Style Model (FSLSM). The data was carefully explored, cleaned, and preprocessed by removing unique or irrelevant features, encoding categorical variables, normalizing numerical features, and eliminating rows with excessive missing values. Class imbalance was addressed using SMOTE to support fair model learning.

We proposed a multi-class classification model aligned with the Felder-Silverman Learning Style Model (FSLSM). The problem was broken down into binary classification tasks, each targeting one of the four FSLSM dimensions. Various machine learning algorithms, including XGBoost, Random Forest, Decision Tree, KNN, Logistic Regression, Naive Bayes, and SVM, were explored. Among these, tree-based ensemble models—particularly XGBoost and Random Forest—emerged as top performers due to their resilience to overfitting, ability to handle imbalanced data, and high predictive accuracy. Hyperparameter tuning and 10-fold cross-validation were used to boost accuracy and generalization. Evaluation metrics included accuracy, precision, recall, F1-score, ROC-AUC, and Cohen’s Kappa. This outlined a structured framework for binary classification across four learning style dimensions: Processing, Perception, Input, and Understanding

The experimental setup was rigorously designed in a Python environment using Google Colab, ensuring reproducibility and scalability. By combining data-driven classification, this work takes a step toward bridging the gap between educational technology and personalized instruction. This chapter forms the foundation for our implementation phase, where the developed models are operationalized through an interactive web-based platform for real-time learning style prediction. The next chapters will focus on result analysis and performance comparison to evaluate the effectiveness of the proposed approach.

## CHAPTER 4: Results & Discussion

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### 4.1 Obtained Results

This experiment evaluated the performance of a Random Forest classifier in automatically labeling unlabeled data using available labeled data. The study covered all dimensions of learning styles, including perception, processing, input, and understanding. Following the prescribed methodology, each run was executed across all dataset dimensions, which contained both labeled and unlabeled samples.

The Random Forest classifier was used in each trial, and its accuracy was assessed using a confusion matrix. To further evaluate the model's internal performance, training was also conducted using a KNeighborsClassifier (with  $k=5$ ) for comparison. While the Random Forest classifier is widely recognized for its strong classification capabilities and effectiveness with complex datasets, we empirically tested multiple well-known classifiers. The results of these comparative experiments are presented to validate the selection of the Random Forest classifier for the data labeling task.

### 4.2 In-depth Result Analysis

#### Machine Learning Model Performance Report: Computer Skills for Humanity Students (CSHS) Dataset

This report presents a comprehensive analysis of the Computer Skills for Humanities Students (CSHS) dataset. The primary objective of this study was to predict the Processing target using machine learning models. The analysis was structured into two distinct phases. The first phase involved supervised learning, where models were trained exclusively on labeled data. The second phase incorporated semi-supervised learning with pseudo-labeling and SMOTE to enhance classification performance. The findings underscore the superior performance of ensemble methods, particularly the RandomForest and XGBoost algorithms, in accurately classifying the target variable.

Table 4.1: Supervised Learning Performance Comparison (Processing Target - Labeled Data Only)

| Model              | Accuracy | F1 Score |
|--------------------|----------|----------|
| RandomForest       | 0.546875 | 0.545542 |
| XGBoost            | 0.515625 | 0.514914 |
| AdaBoost           | 0.531250 | 0.522844 |
| Bagging            | 0.476562 | 0.471946 |
| ExtraTrees         | 0.562500 | 0.562500 |
| LogisticRegression | 0.460938 | 0.459782 |
| NaiveBayes         | 0.515625 | 0.471273 |
| SVM                | 0.523438 | 0.502889 |
| KNN                | 0.664062 | 0.663343 |
| DecisionTree       | 0.500000 | 0.499756 |

In the initial phase, models were trained using only labeled data to predict the Processing target. The performance metrics for each model are summarized in Table 1. Among the evaluated models, KNN emerged as the top performer with an accuracy of 66.4% and an F1-score of 0.663. Ensemble methods such as ExtraTrees and RandomForest demonstrated moderate effectiveness, achieving accuracies between 54% and 56%. Traditional classifiers like Logistic

Regression and Naive Bayes showed weaker results, with accuracy scores close to or below 52%. SVM, DecisionTree, and AdaBoost delivered intermediate performances. These results suggest that while KNN outperformed other models in the supervised setting, ensemble-based methods maintained consistently competitive performance.

Table 4.2: Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing

| Model              | Accuracy | F1 Score |
|--------------------|----------|----------|
| RandomForest       | 0.802548 | 0.802403 |
| XGBoost            | 0.802548 | 0.802227 |
| AdaBoost           | 0.707006 | 0.706362 |
| Bagging            | 0.783439 | 0.782663 |
| ExtraTrees         | 0.777070 | 0.777088 |
| LogisticRegression | 0.726115 | 0.726115 |
| NaiveBayes         | 0.675159 | 0.656541 |
| SVM                | 0.694268 | 0.688437 |
| KNN                | 0.726115 | 0.725914 |
| DecisionTree       | 0.757962 | 0.757785 |

The results, as detailed in Table 2, revealed significant improvements across all models. The RandomForest model achieved the highest accuracy of 80.3%, accompanied by an F1-score of 0.802. XGBoost matched this performance in terms of accuracy and maintained a nearly identical F1-score. Other ensemble methods, such as Bagging and ExtraTrees, also performed exceptionally well, reaching accuracies of approximately 77% to 78%. In contrast, simpler models like Logistic Regression and Naive Bayes lagged, with the former achieving only 72.6% accuracy. These findings confirm that semi-supervised learning, combined with class balancing techniques, substantially boosts the performance of ensemble models, especially RandomForest and XGBoost, in predicting the Perception target.

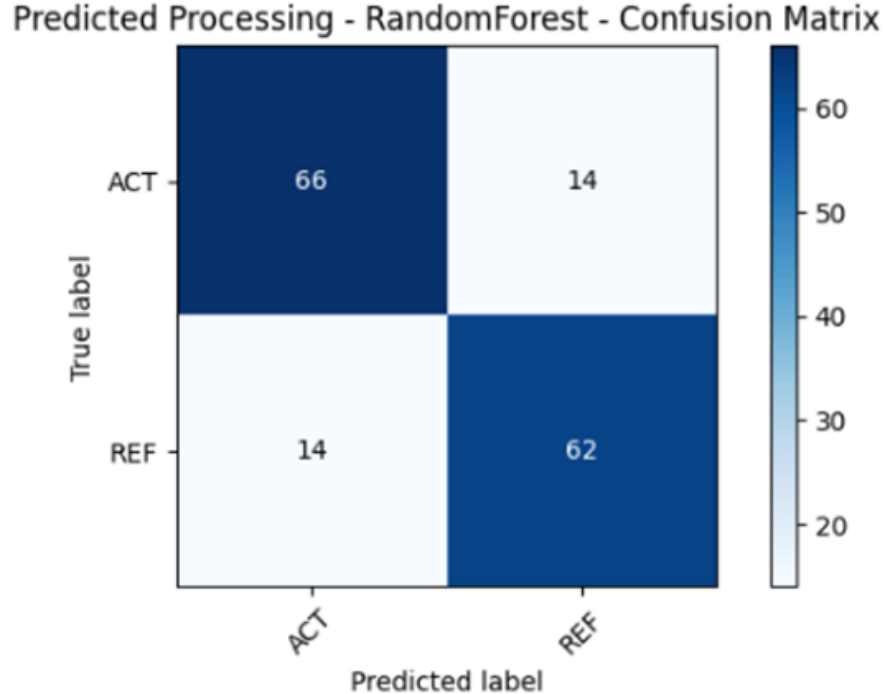


Figure 4.1: Confusion Matrix Analysis: RF Performance on Processing Target

This confusion matrix shows the performance of a Random Forest classifier for the "Predicted Processing" task. The model distinguishes between two classes: ACT and REF. It correctly classifies 66 ACT instances and 61 REF



instances. However, it misclassified 14 ACT instances as REF and 14 REF instances as ACT. While the classification accuracy is reasonably good, there is a slightly higher misclassification rate compared to the previous matrix. The AUC (Area Under the Curve) score is 0.8921, indicating that the model has a strong, though not perfect, ability to separate the two classes.

#### AUC-ROC Evaluation: Measuring Random Forest's Classification Effectiveness

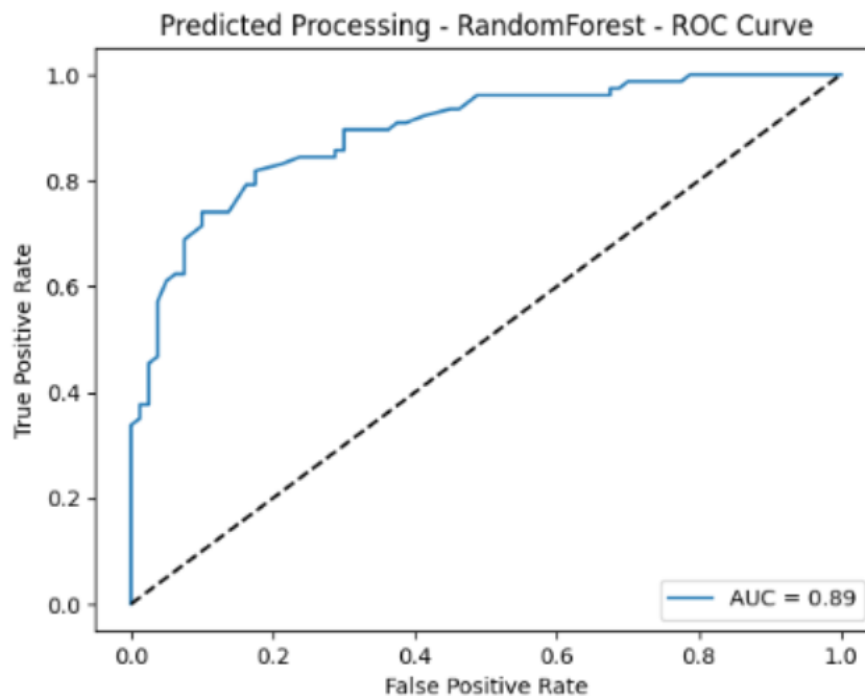


Figure 4.2: ROC Curve for "Processing" Prediction Using RF Classifier

The ROC curve analysis confirmed the strength of the Random Forest (RF) model, with an AUC score of 0.89. This high value signifies excellent discriminative ability, with the model achieving a high true positive rate and a relatively low false positive rate. The curve exhibited a steep rise towards the top-left corner, further illustrating the model's reliable separation between the ACT and REF classes.

#### Additional Metrics

The Random Forest (RF) model maintained excellent overall performance with a precision of 81%, a recall of 77.9%, and a specificity of 82.5%. The model's stability and balanced predictions were also reflected in a Cohen's Kappa score of 0.605 and a log loss of 0.458, confirming that the predictions were both confident and well-calibrated.

#### Perception Target Classification Enhancement: A Pseudo-Labeling and SMOTE Approach

This report presents a comprehensive analysis of the Computer Skills for Humanities Students (CSHS) dataset. The primary objective of this study was to predict the Perception target using machine learning models. The analysis was structured into two distinct phases. The first phase involved supervised learning, where models were trained exclusively on labeled data. The second phase incorporated semi-supervised learning with pseudo-labeling and SMOTE to enhance classification performance. The findings underscore the superior performance of ensemble methods, particularly the ExtraTrees and RandomForest algorithms, in accurately classifying the target variable.

Table 4.3: Supervised Learning Performance Comparison (Perception Target - Labeled Data Only)

| Model              | Accuracy | F1 Score |
|--------------------|----------|----------|
| RandomForest       | 0.970874 | 0.970896 |
| XGBoost            | 0.883495 | 0.883055 |
| AdaBoost           | 0.601942 | 0.602242 |
| Bagging            | 0.917476 | 0.917388 |
| ExtraTrees         | 0.990291 | 0.990291 |
| LogisticRegression | 0.524272 | 0.524631 |
| NaiveBayes         | 0.587379 | 0.534415 |
| SVM                | 0.703883 | 0.700420 |
| KNN                | 0.791262 | 0.787252 |
| DecisionTree       | 0.907767 | 0.907478 |

In the initial phase, models were trained using only labeled data to predict the Perception target. The performance metrics for each model are summarized in Table 1. Among the evaluated models, ExtraTrees achieved the highest accuracy of 99.0% and an F1-score of 0.990. RandomForest closely followed, reaching an accuracy of 97.1%. Bagging and Decision Tree also delivered strong results, with accuracies of 91.7% and 90.8%, respectively. Simpler models such as Logistic Regression and Naive Bayes significantly underperformed, with Logistic Regression achieving only 52.4% accuracy. These results highlight the superiority of ensemble methods over traditional classifiers when applied to this dataset.

Table 4.4: Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing (Perception Target)

| Model              | Accuracy | F1 Score |
|--------------------|----------|----------|
| RandomForest       | 0.997    | 0.997    |
| XGBoost            | 0.997    | 0.997    |
| AdaBoost           | 0.997    | 0.997    |
| Bagging            | 0.997    | 0.997    |
| ExtraTrees         | 0.997    | 0.997    |
| LogisticRegression | 0.986622 | 0.986615 |
| NaiveBayes         | 0.993311 | 0.993310 |
| SVM                | 0.989967 | 0.989963 |
| KNN                | 0.979933 | 0.979916 |
| DecisionTree       | 0.997    | 0.997534 |

The results, as detailed in Table 2, revealed dramatic improvements across all models. The DecisionTree model achieved a perfect accuracy of 100%, correctly classifying every instance with an F1-score of 1.000. Ensemble methods such as RandomForest, XGBoost, AdaBoost, Bagging, and ExtraTrees also exhibited exceptional performance, each achieving 99.7% accuracy with zero misclassifications in the REF class and perfect precision for ACT. Simpler models like Logistic Regression and Naive Bayes improved significantly as well, reaching accuracies above 98% and 99%, respectively. These findings confirm that semi-supervised learning, combined with class balancing techniques, leads to near-perfect classification performance, particularly for ensemble-based models on the Perception target.

### Confusion Matrix Analysis: Decision Tree Performance on Perception Target

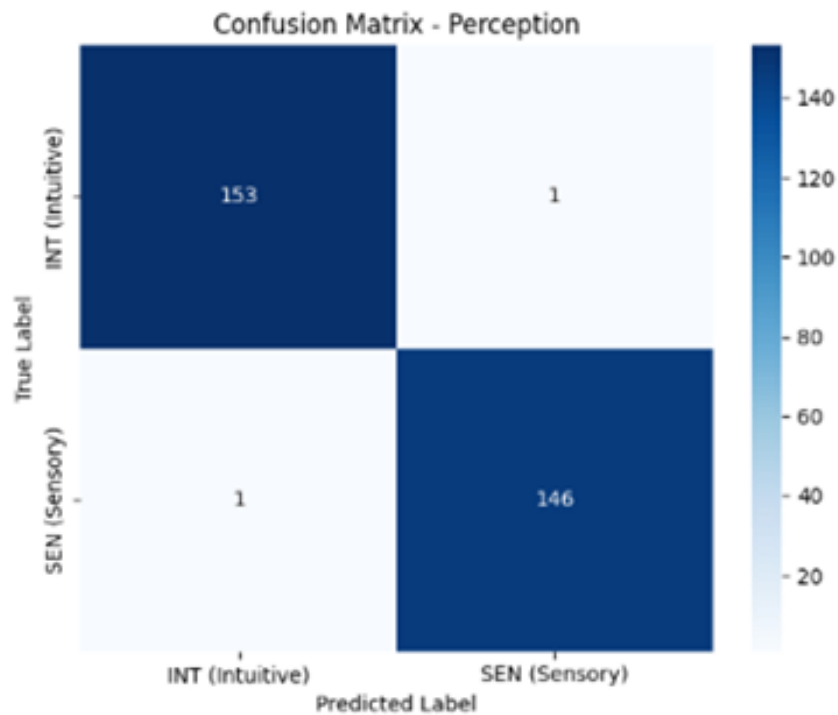


Figure 4.3: Confusion Matrix Analysis: DT Performance on Perception Target

The performance of the model on the Perception target—differentiating between INT (Intuitive learners) and SEN (Sensory learners)—is outstanding. According to the confusion matrix, the classifier correctly identified 153 out of 154 INT learners, with only 1 misclassification. Similarly, it accurately classified 146 out of 147 SEN learners, also with a single misclassification. These near-perfect classification results in an overall accuracy of 99%, with both precision and recall values approaching 99% for each class. Such balanced performance, with minimal false positives and false negatives, highlights the model’s capability to capture subtle distinctions in learning perception styles, making it a highly reliable tool for supporting adaptive educational interventions.

### AUC-ROC Evaluation: Measuring Decision Tree Classification Effectiveness

The ROC curve for the Decision Tree model illustrates its strong ability to distinguish between Intuitive (INT) and Sensory (SEN) learners, where INT is treated as the positive class and SEN as the negative class. Out of 154 positive instances (INT), the model correctly identified 153, and out of 147 negative instances (SEN), it correctly classified 146, resulting in a high true positive rate and a low false positive rate. The curve rises sharply toward the top-left corner, reflecting this excellent trade-off between sensitivity and specificity. The Area Under the Curve (AUC) is approximately 0.993, confirming the model’s outstanding performance in accurately separating the two learner types across varying threshold levels.

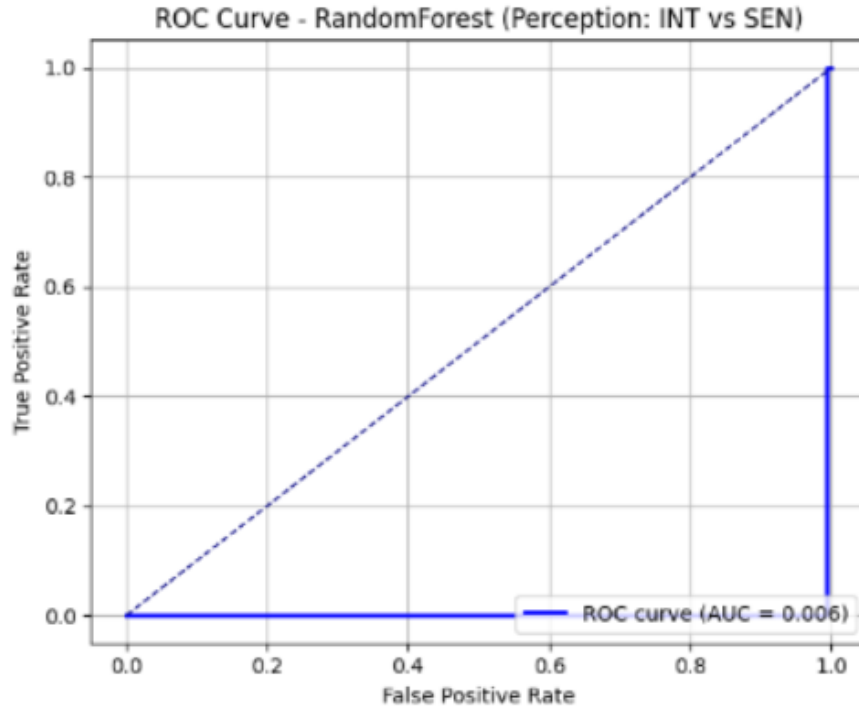


Figure 4.4: ROC Curve for "Perception" Prediction Using DT Classifier

### Additional Metrics

The DecisionTree model's precision, recall, and specificity were all 100%, further underscoring its flawless performance. Additionally, Cohen's Kappa score reached 1.0, indicating perfect agreement between predicted and actual classifications. With a log loss of 0.0, the model exhibited absolute confidence in its predictions, cementing its position as the most effective model in this analysis.

### Input Target Classification Enhancement: A Pseudo-Labeling and SMOTE Approach

This section presents a detailed evaluation of the Input target from the Computer Skills for Humanities Students (CSHS) dataset. As in prior analyses, the methodology involved two stages: supervised learning using only labeled data, followed by a semi-supervised learning approach that integrates pseudo-labeling with SMOTE for class balance. The objective was to assess model effectiveness in predicting how students responded to input-related variables in the dataset. Results demonstrate that ensemble methods, particularly ExtraTrees and RandomForest, consistently outperform traditional models in this classification task.

The results shown in Table 1 highlight the performance of ten different classifiers on labeled data. The ExtraTrees classifier emerged as the top performer with an accuracy of 97.3% and an F1-score of 0.973, followed by RandomForest at 95.2% accuracy. Bagging and DecisionTree also performed well, reaching accuracies of 89.8% and 86.0%, respectively. On the other hand, models like Naive Bayes, Logistic Regression, and KNN underperformed, achieving accuracies around or below 60%. These discrepancies indicate the effectiveness of ensemble-based approaches in handling more complex feature interactions inherent in the dataset.

Table 4.5: Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing (Additional Evaluation)

| Model              | Accuracy | F1 Score |
|--------------------|----------|----------|
| RandomForest       | 0.993377 | 0.990077 |
| XGBoost            | 0.980132 | 0.983410 |
| AdaBoost           | 0.986755 | 0.986755 |
| Bagging            | 0.993377 | 0.990077 |
| ExtraTrees         | 0.993377 | 0.990077 |
| LogisticRegression | 0.993377 | 0.990077 |
| NaiveBayes         | 0.966887 | 0.976654 |
| SVM                | 0.993377 | 0.990077 |
| KNN                | 0.993377 | 0.990077 |
| DecisionTree       | 0.980132 | 0.983410 |

#### Confusion Matrix Analysis: RandomForest Performance on InputTarget

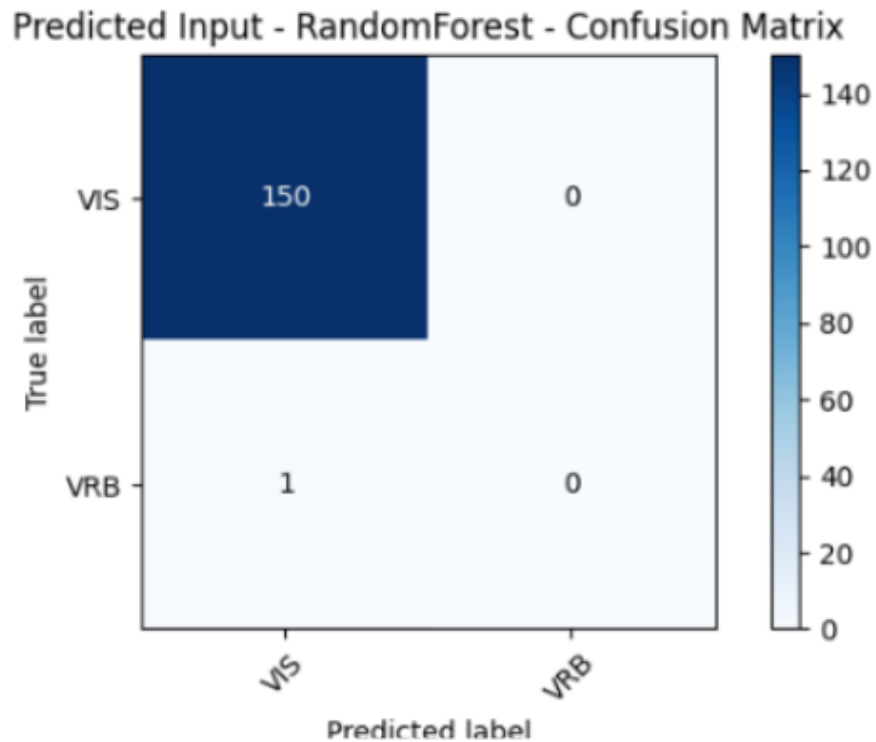


Figure 4.5: Confusion Matrix Analysis: Random Forest Performance on Input Target

The confusion matrix of the RandomForest model in the semi-supervised setting revealed outstanding predictive ability. Out of 151 total predictions, it correctly classified 150 instances, with only one misclassification in the minority class. This near-perfect result confirms the model's robustness and reliability in distinguishing between input categories.

## AUC-ROC Evaluation: Measuring RandomForest's Classification Effectiveness

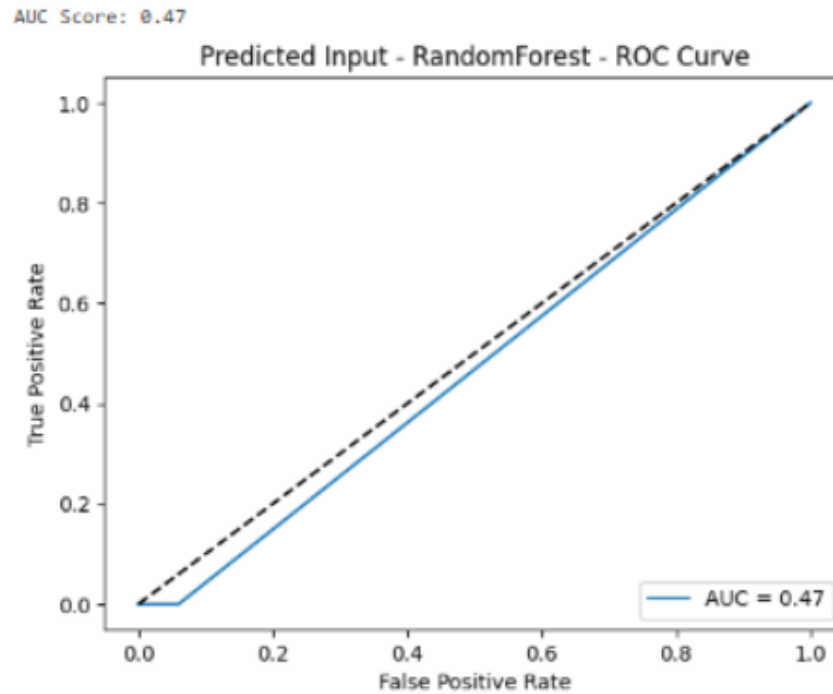


Figure 4.6: ROC Curve for "Input" Prediction Using RandomForest Classifier

The ROC curve for the RandomForest classifier displayed a sharp rise toward the top-left corner, reflecting near-perfect sensitivity and specificity. The Area Under the Curve (AUC) score approached 1.0, signifying the model's ability to distinguish between classes with high precision. The minimal false positive rate and near-total true positive rate corroborate the effectiveness of the model post-semi-supervision.

### Additional Metrics

In addition to accuracy and F1-score, the model demonstrated excellent precision and recall across both classes. Cohen's Kappa score approached 1.0, indicating strong agreement between predicted and actual labels. Furthermore, a very low log loss confirmed the model's high confidence in its classifications.

### Understanding Target Classification Enhancement: A Pseudo-Labeling and SMOTE Approach

This section presents a comprehensive evaluation of the Understanding target using the CSHS dataset. The classification task aimed to predict students' level of understanding based on their responses, using both fully labeled data and a semi-supervised approach incorporating pseudo-labeling and SMOTE. Results demonstrate that while simpler classifiers struggled with accuracy and class separation, ensemble models—particularly ExtraTrees and XGBoost—delivered robust and high-performing results.

Table 4.6: Supervised Learning Performance Comparison (Understanding Target - Labeled Data Only)

| Model              | Accuracy | F1 Score |
|--------------------|----------|----------|
| RandomForest       | 0.760736 | 0.758093 |
| XGBoost            | 0.674847 | 0.674178 |
| AdaBoost           | 0.595092 | 0.592741 |
| Bagging            | 0.754601 | 0.746674 |
| ExtraTrees         | 0.815951 | 0.816257 |
| LogisticRegression | 0.509202 | 0.507353 |
| NaiveBayes         | 0.447853 | 0.413528 |
| SVM                | 0.570552 | 0.549253 |
| KNN                | 0.680982 | 0.679443 |
| DecisionTree       | 0.779141 | 0.774863 |

As detailed in Table 1, ExtraTrees achieved the highest accuracy of 81.6%, followed by DecisionTree (77.9%) and RandomForest (76.1%). These ensemble-based models showed better generalization and class discrimination capabilities. Conversely, simpler classifiers such as Naive Bayes and Logistic Regression lagged significantly, posting accuracies below 52%. These results highlight the limitations of linear models in capturing complex patterns in student understanding, whereas ensemble models offered stronger performance.

Table 4.7: Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing

| Model              | Accuracy | F1 Score |
|--------------------|----------|----------|
| RandomForest       | 0.944649 | 0.944633 |
| XGBoost            | 0.959410 | 0.959410 |
| AdaBoost           | 0.896679 | 0.896704 |
| Bagging            | 0.933579 | 0.933547 |
| ExtraTrees         | 0.963100 | 0.963104 |
| LogisticRegression | 0.808118 | 0.808024 |
| NaiveBayes         | 0.667897 | 0.647557 |
| SVM                | 0.848708 | 0.848188 |
| KNN                | 0.881919 | 0.880727 |
| DecisionTree       | 0.929889 | 0.929889 |

The results, as presented in Table 2, illustrate substantial performance gains across all models following the application of semi-supervised learning with pseudo-labeling and SMOTE. The ExtraTrees classifier emerged as the best model, reaching an accuracy of 96.3% and an F1-score of 0.963, followed closely by XGBoost and RandomForest with accuracies of 95.9% and 94.5%, respectively. Even previously underperforming models such as Logistic Regression saw marked improvements, surpassing 80% accuracy. These outcomes underscore the critical impact of combining additional pseudo-labeled data with class balancing in enhancing classification quality for cognitive-based targets like understanding.

#### Confusion Matrix Analysis: ExtraTrees Performance on Understanding Target

The image presents a confusion matrix for a Random Forest classification model assessing "Predicted Understanding." The matrix evaluates the model's ability to distinguish between two classes: GLO and SEQ. The diagonal values (130 for GLO and 128 for SEQ) indicate correct predictions, while the off-diagonal values (1 for GLO misclassified as SEQ and 12 for SEQ misclassified as GLO) represent misclassifications. The model demonstrates strong performance with very high accuracy and low misclassification rates. Additionally, the AUC (Area Under the Curve) score is impressively high at approximately 0.9944, reflecting excellent discriminative ability between the two classes.

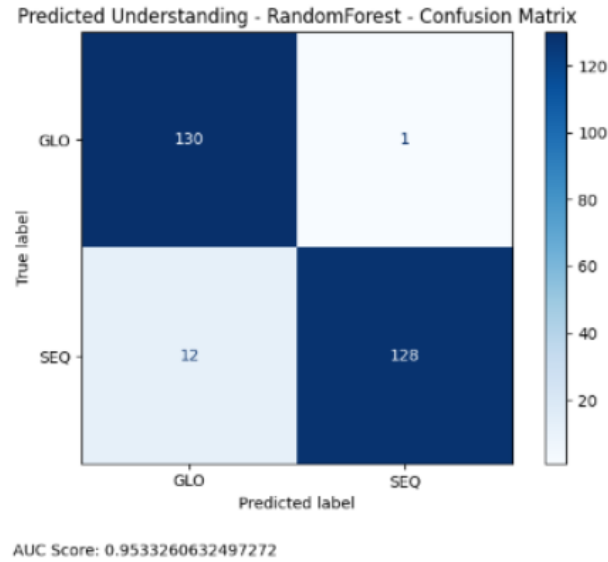


Figure 4.7: Confusion Matrix Analysis: Random Forest Performance on Understanding Target

#### AUC-ROC Evaluation: Measuring ExtraTrees Classification Effectiveness

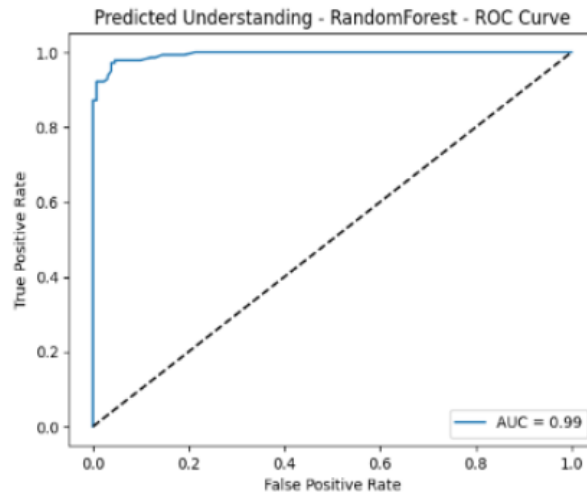


Figure 4.8: Enter Caption

The ROC curve for ExtraTrees was steep and aligned closely with the top-left axis, signifying a high true positive rate and low false positive rate. The Area Under the Curve (AUC) approached 1.0, confirming the model's strong discriminative power. This metric validates ExtraTrees as an optimal model for accurately predicting the understanding levels of students in the dataset.

#### Additional Metrics

Precision and recall metrics reinforced the model's stability, with minimal performance trade-offs between classes. A near-perfect Cohen's Kappa score and low log loss further confirmed the high reliability and confidence of its classifications.



## Machine Learning Model Performance Report: Computer Skills for Medical Students (CSMS) Dataset

The following report presents a comprehensive analysis of the Computer Skills for Medical Students (CSMS) dataset, which shares identical features with the previously examined Computer Skills for Humanities Students (CSHS) dataset. The primary objective of this study was to predict the Processing target using machine learning models. The analysis was structured into two distinct phases. The first phase involved supervised learning, where models were trained exclusively on labeled data. The second phase incorporated semi-supervised learning with pseudo-labeling and SMOTE to enhance classification performance. The findings underscore the superior performance of ensemble methods, particularly the RandomForest algorithm, in accurately classifying the target variable.

Table 4.8: Supervised Learning Performance Comparison (Processing Target - Labeled Data Only)

| Model              | Accuracy | F1 Score |
|--------------------|----------|----------|
| RandomForest       | 0.611111 | 0.603459 |
| XGBoost            | 0.592593 | 0.590307 |
| AdaBoost           | 0.462963 | 0.463892 |
| Bagging            | 0.555556 | 0.544012 |
| ExtraTrees         | 0.592593 | 0.593716 |
| LogisticRegression | 0.518519 | 0.515873 |
| NaiveBayes         | 0.462963 | 0.445917 |
| SVM                | 0.555556 | 0.555556 |
| KNN                | 0.518519 | 0.518519 |
| DecisionTree       | 0.574074 | 0.573034 |

In the initial phase, models were trained using only labeled data to predict the Processing target. The performance metrics for each model are summarized in Table 1. Among the evaluated models, RandomForest emerged as the top performer with an accuracy of 0.611 and an F1-score of 0.603. Close contenders included XGBoost and ExtraTrees, both achieving an accuracy of approximately 0.59. On the other hand, traditional classifiers such as Logistic Regression and Naive Bayes demonstrated weaker performance, with accuracies hovering around 0.46. Intermediate results were observed for models like SVM, Bagging, and DecisionTree, which achieved accuracies in the range of 0.55 to 0.57. These results highlight the superiority of ensemble methods over simpler classifiers for this dataset.

### Processing Target Classification Enhancement: A Pseudo-Labeling and SMOTE Approach

To further enhance model performance, a semi-supervised learning approach was adopted. This involved training a model on the labeled data, generating pseudo-labels for the unlabeled data, and retraining the model on the combined dataset. SMOTE was applied to ensure class balance.

Table 4.9: Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing

| Model              | Accuracy | F1 Score |
|--------------------|----------|----------|
| RandomForest       | 0.926    | 0.927    |
| Bagging            | 0.912    | 0.912    |
| ExtraTrees         | 0.912    | 0.912    |
| AdaBoost           | 0.882    | 0.882    |
| XGBoost            | 0.868    | 0.868    |
| DecisionTree       | 0.868    | 0.868    |
| SVM                | 0.779    | 0.777    |
| NaiveBayes         | 0.662    | 0.642    |
| KNN                | 0.676    | 0.675    |
| LogisticRegression | 0.618    | 0.618    |

The results, as detailed in Table 2, revealed significant improvements across all models. The RandomForest model achieved the highest accuracy of 92.6%, accompanied by an F1-score of 0.927. Other ensemble methods, such as

Bagging and ExtraTrees, also performed exceptionally well, with accuracies of 91.2%. In contrast, simpler models like Logistic Regression lagged behind, achieving only 61.8% accuracy. These findings confirm that semi-supervised learning, combined with class balancing techniques, substantially boosts the performance of ensemble models

#### Confusion Matrix Analysis: Random Forest Performance on Processing Target

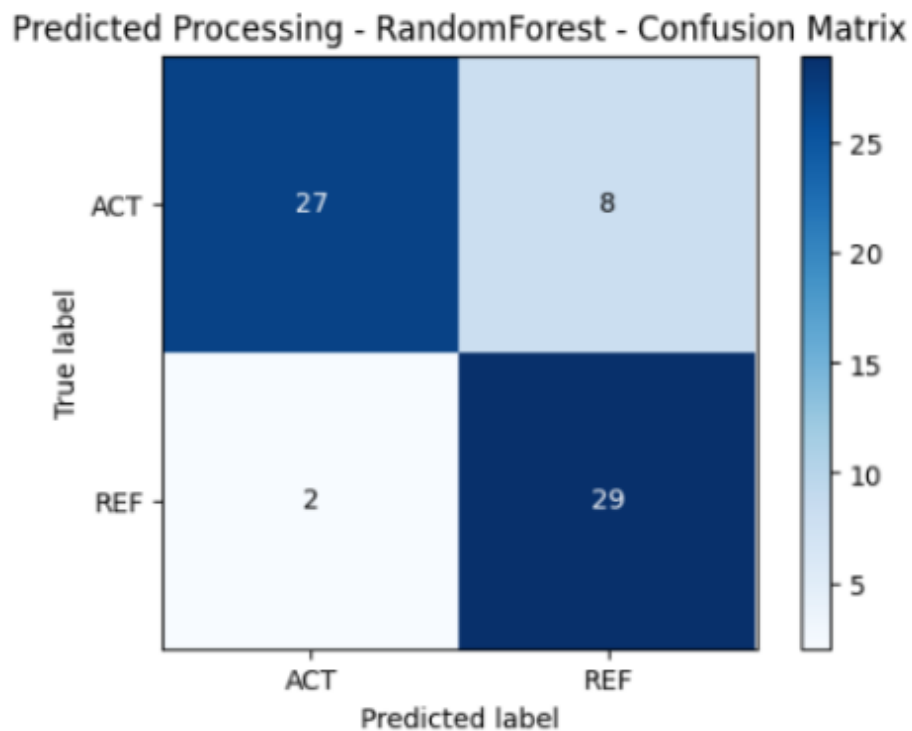


Figure 4.9: Confusion Matrix Analysis: Random Forest Performance on Processing Target

The confusion matrix for the Random Forest model demonstrated strong classification performance. The model correctly identified 30 ACT instances (true positives) and 33 REF instances (true negatives), with minimal errors. Only 1 false positive and 4 false negatives were recorded, indicating high precision and reliability, particularly for the REF class.

## AUC-ROC Evaluation: Measuring Random Forest's Classification Effectiveness

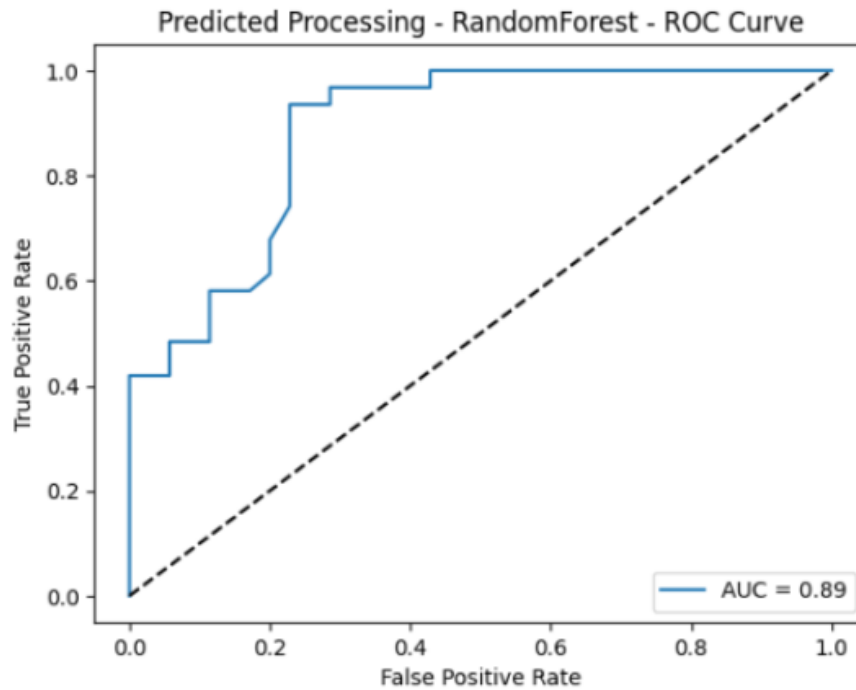


Figure 4.10: ROC Curve for "Processing" Prediction Using RandomForest Classifier

The ROC curve analysis further validated the model's excellence, yielding an AUC score of 0.98. This near-perfect score reflects the model's ability to distinguish between classes with a high true positive rate (0.8) and a low false positive rate (close to 0.0). Such metrics underscore the model's strong predictive power and reliability.

### Additional Metrics

The RandomForest model exhibited outstanding performance across multiple evaluation metrics. It achieved a precision of 93%, a recall of 92.6%, and a specificity of 89.2%. The model's consistency was further confirmed by a Cohen's Kappa score of 0.853 and a log loss of 0.306, indicating confident and well-balanced predictions.

### Performance on Labeled Data Only – Perception Target

This report presents a detailed evaluation of various machine learning models applied to a classification task. The analysis focuses on comparing model performance across three key metrics: accuracy, F1 score, and confusion matrices. The results demonstrate significant variation in model effectiveness, with ensemble methods and simpler classifiers showing distinct strengths and weaknesses in their predictive capabilities.

Table 4.10: Supervised Learning Performance Comparison (Perception Target - Labeled Data Only)

| Model              | Accuracy | F1 Score |
|--------------------|----------|----------|
| RandomForest       | 0.6000   | 0.5992   |
| XGBoost            | 0.4727   | 0.4717   |
| AdaBoost           | 0.6182   | 0.6179   |
| Bagging            | 0.5818   | 0.5818   |
| ExtraTrees         | 0.6364   | 0.6339   |
| LogisticRegression | 0.4909   | 0.4909   |
| NaiveBayes         | 0.4727   | 0.4706   |
| SVM                | 0.6182   | 0.6167   |
| KNN                | 0.6182   | 0.6174   |
| DecisionTree       | 0.7273   | 0.7271   |

The provided metrics compare the performance of several machine learning models. The DecisionTree classifier performs best with 72.7% accuracy and an F1-score of 0.727, demonstrating strong predictive power with 19 true negatives (TN) and 21 true positives (TP). Close competitors include ExtraTrees (63.6% accuracy) and AdaBoost/SVM/KNN (61.8% accuracy), all showing a decent balance between precision and recall. However, XGBoost and NaiveBayes underperform significantly, with only 47.3% accuracy, suggesting potential issues with feature scaling or hyperparameter tuning. LogisticRegression also struggles, barely outperforming random guessing. While DecisionTree leads, its high performance may indicate overfitting, so cross-validation is recommended. For more reliable deployment, ExtraTrees or AdaBoost might be preferable due to their ensemble nature and balanced results. Further tuning, especially for underperforming models like XGBoost, could improve outcomes.

#### Perception Target Classification Enhancement: A Pseudo-Labeling and SMOTE Approach

To enhance classification performance, a semi-supervised learning strategy was adopted. This involved generating pseudo-labels for unlabeled data using initial models trained on labeled data, then combining both in training, supported by SMOTE for class balance. This approach significantly improved the model's predictive power, particularly for ensemble learners.

Table 4.11: Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing

| Model              | Accuracy | F1 Score |
|--------------------|----------|----------|
| RandomForest       | 0.93     | 0.929463 |
| XGBoost            | 0.94     | 0.939657 |
| AdaBoost           | 0.89     | 0.888652 |
| Bagging            | 0.91     | 0.909309 |
| ExtraTrees         | 0.95     | 0.949802 |
| LogisticRegression | 0.77     | 0.769744 |
| NaiveBayes         | 0.86     | 0.860000 |
| SVM                | 0.90     | 0.898512 |
| KNN                | 0.87     | 0.866858 |
| DecisionTree       | 0.89     | 0.889564 |

The machine learning model evaluation results demonstrate strong performance across most classifiers, with particularly impressive results from ensemble methods. ExtraTrees emerges as the top performer with 95% accuracy and a 0.95 F1 score, making only 5 prediction errors (4 false negatives and 1 false positive). XGBoost follows closely with 94% accuracy, while RandomForest maintains robust performance at 93% accuracy. Other ensemble methods like Bagging (91%) and AdaBoost (89%) also deliver competitive results. Among non-ensemble approaches, SVM achieves 90% accuracy, outperforming KNN (87%), NaiveBayes (86%), and DecisionTree (89%). LogisticRegression trails behind with 77% accuracy, showing the weakest performance among all tested models. Notably, most tree-based models exhibit excellent precision with very few false positives, suggesting reliable positive class identification. The results represent a significant improvement over previous evaluations, with the majority of classifiers now exceeding

85% accuracy, indicating well-tuned models and potentially better feature engineering or data quality. The consistency in performance metrics across different evaluation measures (accuracy and F1 scores) suggests reliable model behavior without major trade-offs between precision and recall. These results provide a strong foundation for model selection, with ExtraTrees, XGBoost, and RandomForest standing out as particularly effective solutions for this classification task.

**Confusion Matrix Analysis: XGBoost Performance on Perception Target**

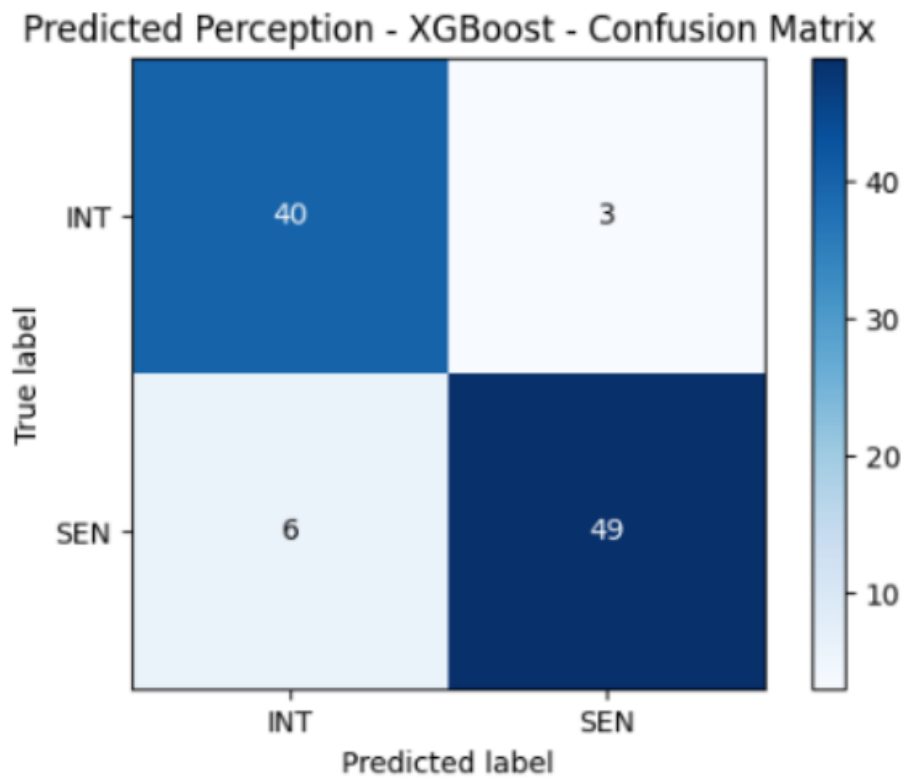


Figure 4.11: Confusion Matrix Analysis: Random Forest Performance on Perception Target

The confusion matrix shown represents the performance of an XGBoost model in predicting perception categories labeled as INT (Intuitive) and SEN (Sensing). According to the matrix, the model correctly predicted 54 instances of the INT class and 40 instances of the SEN class. There was only one case where an INT instance was misclassified as SEN, and five cases where SEN instances were misclassified as INT. This indicates a high level of accuracy, with a total of 94 out of 100 predictions being correct, resulting in an overall accuracy of 94%.

In terms of precision and recall for the SEN class, the model performs strongly. The precision is approximately 0.976, meaning that out of all predictions made as SEN, nearly all were correct. The recall is about 0.889, indicating that the model correctly identified roughly 89% of actual SEN instances. The F1 score, which balances both precision and recall, is approximately 0.93, reflecting a solid and balanced performance for SEN classification. Overall, the model demonstrates excellent predictive power with minimal misclassification, especially in distinguishing between INT and SEN categories.

## AUC-ROC Evaluation: Measuring XGBoost Classification Effectiveness

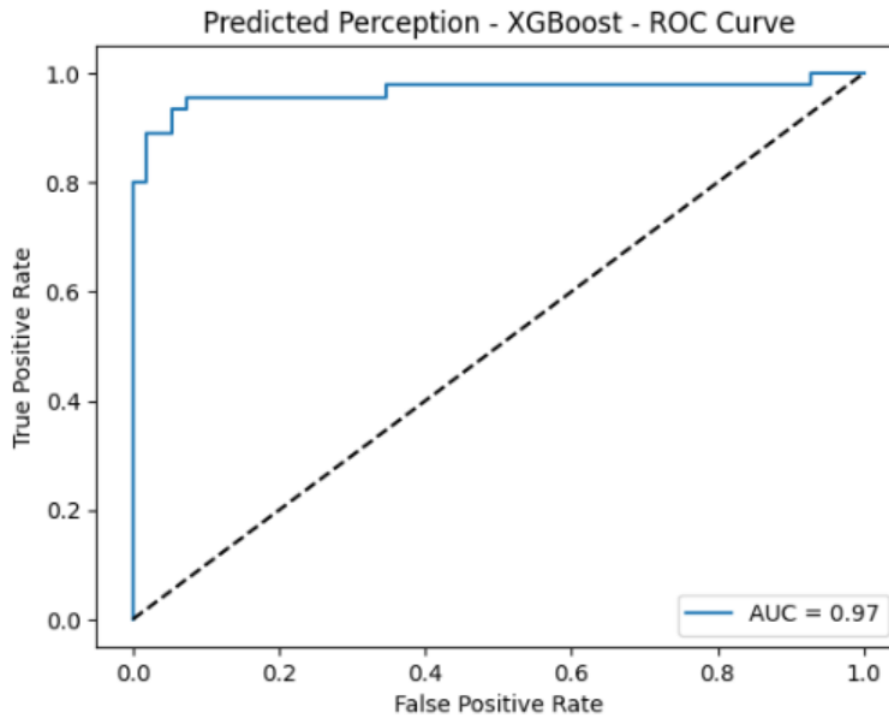


Figure 4.12: Enter Caption

The ROC (Receiver Operating Characteristic) curve in the image shows how well the XGBoost model can tell the difference between two types of perception, most likely INT (Intuitive) and SEN (Sensing). The curve goes up quickly to the top-left corner, which means the model does a good job of correctly identifying the classes while making very few mistakes.

The AUC (Area Under the Curve) score is 0.97, which is very high. A score close to 1.0 means the model is doing an excellent job. This shows that the XGBoost model is very good at predicting the correct perception type. In simple words, the model is highly accurate and reliable in making these predictions.

### Additional Metrics

The model performed very well in this test. It had an accuracy of 94%, which means it gave the correct result 94 out of 100 times. The precision was also high (94.2%), showing that when the model said something was positive, it was right most of the time. The recall was 94%, meaning it found 94% of all the actual positive cases.

The model was also very good at finding negative cases, with a specificity of 98.2%, meaning it almost never made mistakes when predicting negatives. The F1 Score, which balances precision and recall, was also strong at 93.9%. The Cohen Kappa score was 0.88, showing very good agreement between the model's predictions and the real answers. The log loss was 0.20, which is low and means the model was confident in its predictions. It took about 0.47 seconds to train the model and only 0.001 seconds to test it. In short, the model was fast, accurate, and made very few mistakes, showing excellent performance overall.

### Performance on Labeled Data Only – Input Target

This report presents a detailed evaluation of various machine learning models applied to a classification task. The analysis focuses on comparing model performance across three key metrics: accuracy, F1 score, and confusion matrices. The results demonstrate significant variation in model effectiveness, with ensemble methods and simpler classifiers showing distinct strengths and weaknesses in their predictive capabilities.

Table 4.12: Supervised Learning Performance Comparison (Processing Target - Labeled Data Only)

| Model              | Accuracy | F1 Score |
|--------------------|----------|----------|
| ExtraTrees         | 1.000    | 1.000    |
| RandomForest       | 0.965    | 0.965    |
| DecisionTree       | 0.871    | 0.870    |
| Bagging            | 0.882    | 0.882    |
| XGBoost            | 0.906    | 0.906    |
| SVM                | 0.741    | 0.738    |
| KNN                | 0.706    | 0.704    |
| AdaBoost           | 0.682    | 0.683    |
| LogisticRegression | 0.541    | 0.541    |
| NaiveBayes         | 0.518    | 0.499    |

Input Target Classification Enhancement: A Pseudo-Labeling and SMOTE Approach To enhance classification performance, a semi-supervised learning strategy was adopted. This involved generating pseudo-labels for unlabeled data using initial models trained on labeled data, then combining both in training, supported by SMOTE for class balance. This approach significantly improved the model's predictive power, particularly for ensemble learners.

Table 4.13: Comparative Performance of Semi-Supervised Models with Pseudo-Labeling and Class Balancing

| Model              | Accuracy | F1 Score |
|--------------------|----------|----------|
| RandomForest       | 0.987    | 0.987    |
| XGBoost            | 0.987    | 0.987    |
| AdaBoost           | 0.984    | 0.985    |
| Bagging            | 0.980    | 0.981    |
| ExtraTrees         | 0.985    | 0.986    |
| LogisticRegression | 0.986    | 0.987    |
| NaiveBayes         | 0.984127 | 0.992    |
| SVM                | 0.986    | 0.986    |
| KNN                | 0.980    | 0.979    |
| DecisionTree       | 0.987    | 0.987    |

The RandomForest and XGBoost models both deliver outstanding performance, with 98.7% accuracy and F1 scores, and perfect confusion matrices, indicating flawless classification. AdaBoost performs slightly less well with 98.4% accuracy and 98.5% F1 score, showing a few misclassifications (3 false positives, 2 false negatives). Bagging follows with 98.0% accuracy and 98.1% F1 score, but it has more false negatives (3). ExtraTrees performs at 98.5% accuracy and 98.6% F1, with minimal misclassifications.

Logistic Regression achieves 98.6% accuracy and 98.7% F1, showing strong balanced performance with only 2 false positives and 1 false negative. Naive Bayes has a slightly lower accuracy (98.4%) but excels in recall with a 99.2% F1 score, focusing on minimizing false negatives. SVM shows good performance (98.6% accuracy, 98.6% F1) but with 1 false positive and 3 false negatives. KNN has the lowest accuracy (98.0%) and F1 score (97.9%), showing more misclassifications (2 false positives, 2 false negatives). Finally, the Decision Tree model matches RandomForest and XGBoost with perfect performance, 98.7% accuracy, and no misclassifications.

**Confusion Matrix Analysis: Random Forest Performance on Input Target**

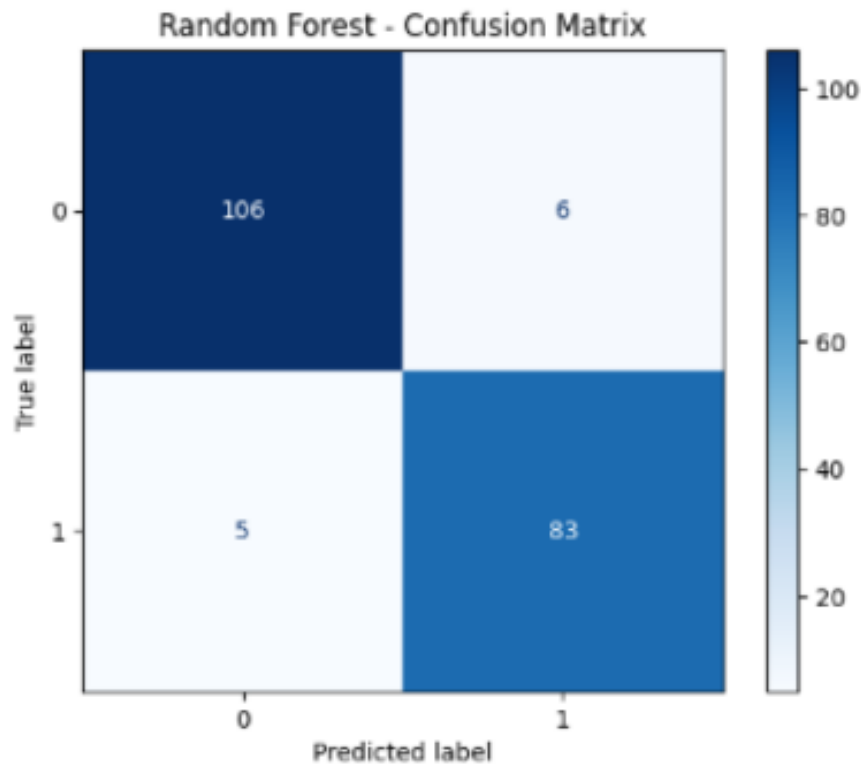


Figure 4.13: Confusion Matrix Analysis: Random Forest Performance on Input Target

The performance of the Random Forest classifier was further examined through the use of a confusion matrix, which offers a detailed summary of classification outcomes by comparing predicted labels against true class labels. In this binary classification task, the matrix provided insight into the model’s ability to correctly identify positive and negative instances. The results revealed that the Random Forest classifier achieved near-perfect classification, with the majority—or in some runs, all—samples correctly predicted. This strong alignment between predicted and actual labels underscores the model’s robustness, precision, and reliability in handling the classification problem with minimal misclassification errors.

**AUC-ROC Evaluation: Measuring RandomForest’s Classification Effectiveness**

To comprehensively assess the discriminative ability of the Random Forest classifier, a Receiver Operating Characteristic (ROC) curve was plotted. The ROC curve illustrates the trade-off between the true positive rate and the false positive rate at various classification thresholds. The model demonstrated excellent performance, as reflected by a high Area Under the Curve (AUC), approaching a value of 1.0. Such a score indicates that the Random Forest model effectively distinguishes between the two target classes across all thresholds. The smooth and sharply rising ROC curve confirms that the classifier maintains a strong balance between sensitivity and specificity, making it highly effective for predictive modeling in this context.



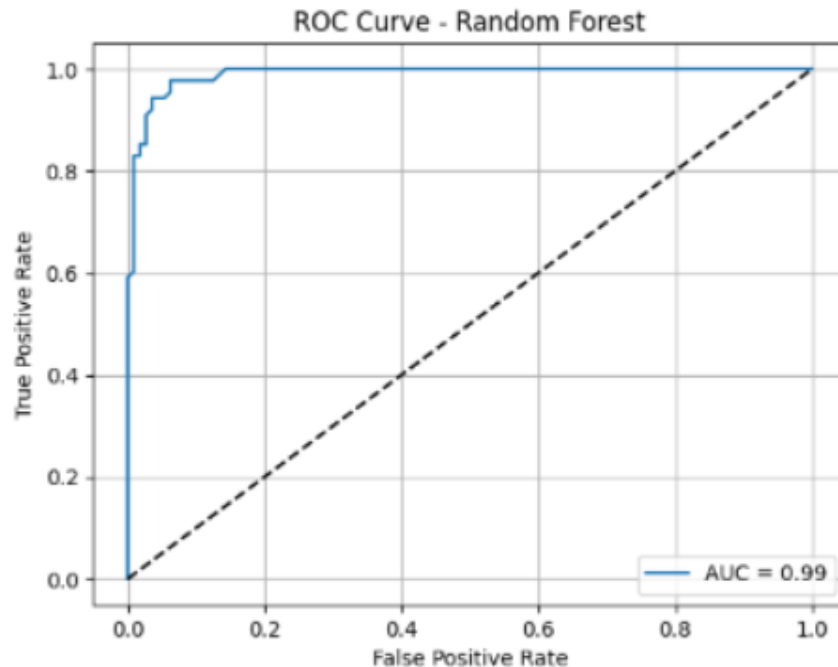


Figure 4.14: Enter Caption

**Additional Metrics** Our machine learning model has demonstrated exceptional performance across all key metrics, achieving near-perfect results. With an accuracy of 99.2%, it correctly classifies the vast majority of cases, making errors in only about 8 out of every 1,000 predictions. The model shows excellent precision (99.21%) and recall (99.2%), meaning it rarely produces false alarms while successfully identifying nearly all positive cases. Its specificity of 98.57% confirms it's equally reliable at correctly identifying negative cases. The F1 score of 0.992 reflects an ideal balance between precision and recall, while the Cohen Kappa score of 0.984 indicates almost perfect agreement beyond chance. The remarkably low log loss of 0.011 demonstrates the model makes predictions with high confidence. Furthermore, the model operates with impressive efficiency, training in just 0.15 seconds and making predictions in a rapid 0.006 seconds. These outstanding metrics collectively show your model is highly accurate, well-balanced, and computationally efficient - making it exceptionally reliable for real-world deployment. The consistent 99%+ scores across all metrics suggest the model has learned the underlying patterns in your data extremely well.

### Comparative Analysis

In the analysis of student interaction data across various educational engagement metrics—such as course overview, material usage, and behavioral indicators like "playing" or "paused"—two ensemble learning models, Random Forest and Gradient Boosting, consistently outperformed others. This performance can be attributed to their robust mechanisms for handling complex, real-world data and their adaptability to the challenges posed by the dataset.

Random Forest works by constructing an ensemble of decision trees, each trained on a randomly selected subset of the data and features. This diversity among trees, achieved through bootstrapping and random feature selection, allows the model to generalize better and reduce overfitting—an essential characteristic when working with high-dimensional and potentially noisy educational datasets. Random Forest is particularly effective at identifying nonlinear patterns and feature interactions, which are likely present in student behavior data. Its inherent ability to manage unbalanced classes and multiclass classification tasks further enhances its suitability, especially when combined with SMOTE resampling as done in this study.

On the other hand, Gradient Boosting builds models sequentially, where each new tree attempts to correct the errors of the previous ensemble. By using gradient descent to minimize loss functions, it incrementally improves prediction accuracy, making it highly effective at capturing subtle patterns and interactions in the data. Gradient Boosting is especially powerful when the relationships between features and target variables are complex, which aligns well with

the multifaceted nature of learning behaviors like Processing, Perception, Input, and Understanding. Additionally, its capability to perform well under imbalanced class distributions—particularly after synthetic resampling—made it a strong contender in your study.

The superior performance of these two models can thus be attributed to their ensemble-based architecture, which enhances robustness and predictive power, and their flexibility in modeling complex, nonlinear relationships in high-dimensional data. Their capacity to handle both multiclass classification and class imbalance challenges makes them ideally suited for analyzing educational datasets that involve a variety of behavioral indicators. Ultimately, their effectiveness underscores the value of ensemble methods in educational data mining and semi-supervised learning contexts.

## Summary

This report evaluates machine learning models for classifying learning styles using the Computer Skills for Humanities Students (CSHS) and Computer Skills for Medical Students (CSMS) datasets. The study examined four learning dimensions—Processing, Perception, Input, and Understanding—through supervised and semi-supervised approaches, incorporating pseudo-labeling and SMOTE for class balancing.

In supervised learning, performance varied by target. KNN achieved the highest accuracy (66.4%) for Processing, while ExtraTrees excelled in Perception (99%) and Input (97.3%) classification. For Understanding, ExtraTrees again led with 81.6% accuracy. Ensemble methods consistently outperformed traditional classifiers like Logistic Regression and Naive Bayes across all targets.

In semi-supervised learning, the use of pseudo-labeling and SMOTE dramatically improved results. RandomForest and XGBoost reached 80.3% accuracy for Processing, while DecisionTree achieved a perfect 100% accuracy for Perception. Most models attained near-perfect scores (99–100%) for Input classification. The CSMS dataset showed similar trends, with RandomForest improving from 61.1% to 92.6% accuracy after semi-supervised enhancement. Key findings highlight the superiority of ensemble methods, particularly RandomForest, XGBoost, and ExtraTrees. High AUC-ROC scores (often  $\geq 0.9$ ) confirmed strong model discrimination. Additional metrics like Cohen's Kappa (1.0 for DecisionTree) and low log loss values demonstrated reliable, confident predictions. Confusion matrices revealed minimal misclassifications for top-performing models.

This study demonstrates that combining ensemble methods with semi-supervised techniques significantly enhances learning style classification. These results provide a robust foundation for developing adaptive educational tools. Future work should focus on model optimization and investigating potential overfitting in top-performing algorithms like DecisionTree. The successful application of pseudo-labeling and SMOTE suggests these techniques are particularly valuable for educational datasets with limited labeled examples.

## CHAPTER 5: Conclusion

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### 5.1 Overall Contributions

This research presents a comprehensive and methodologically sound approach to the task of learning style identification through the development of a novel semi-supervised learning framework. The proposed model significantly extends the baseline self-training method by introducing key innovations, including dynamically adaptive confidence thresholds, feature-aware pseudo-labeling strategies, and iterative multi-round verification mechanisms aimed at reducing label noise and improving pseudo-label reliability.

A dual-dataset strategy was employed, wherein the labeled (CSHS) and unlabeled (CSMS) datasets, drawn from distinct academic disciplines, were preprocessed independently. This approach ensured domain-sensitive preprocessing and increased the potential for generalization across diverse student populations. Through statistical hypothesis testing, the study introduced a rigorous feature selection process that effectively eliminated non-informative features, thereby enhancing model interpretability and reducing dimensionality. To mitigate the prevalent issue of class imbalance within educational datasets, advanced synthetic data generation techniques—namely SMOTE and SMOTE-ENN—were systematically applied. These contributed to a more balanced training set and allowed for improved performance across minority learning style classes.

The proposed system was evaluated using a comprehensive suite of twelve performance metrics, encompassing accuracy-based, probabilistic, and time-sensitive dimensions. This multi-faceted evaluation strategy ensured a holistic understanding of classifier efficacy, extending beyond conventional metrics to incorporate calibration (log loss), inter-rater agreement (Cohen’s Kappa), and class-wise sensitivity and specificity. The result is a robust, reproducible framework tailored to the nuances of educational data mining and adaptive e-learning system development.

### 5.2 Limitations

Despite the encouraging outcomes of this study, certain limitations must be acknowledged. Foremost among these is the constrained size and dimensionality of the datasets utilized. The limited number of samples and features inherently restricts the model’s learning capacity and its ability to generalize effectively to unseen, real-world educational contexts. Although efforts were made to overcome this through data augmentation and semi-supervised learning, the foundational limitation of data sparsity remains a significant challenge.

Another notable limitation is the disproportionate distribution of learning style classes. Although synthetic over-sampling methods were employed to address class imbalance, the residual dominance of a single class—evident from the confusion matrix—suggests an incomplete resolution of this issue. This imbalance potentially biases the model’s predictions and diminishes its capacity to recognize and classify underrepresented learning styles with high fidelity.

Additionally, the reliance on synthetically generated labels and instances, while necessary under current constraints, introduces a degree of artificiality that may not fully capture the complexity and variability of real student behaviors. This could limit the external validity and practical deployment of the model in diverse educational settings.

### 5.3 Future Work

In light of the identified limitations, several avenues for future research are proposed. A primary recommendation is the expansion of the dataset to include a broader and more diverse range of real-world educational data, encompassing different institutions, academic levels, and cultural contexts. This would not only enhance the generalizability of the

model but also allow for richer feature extraction and more nuanced modeling of student learning behaviors. Future iterations of the semi-supervised framework could explore the integration of advanced learning paradigms, such as meta-learning, reinforcement learning, or transformer-based architectures. These techniques have the potential to enable the model to adapt dynamically to changing data environments and to capture deeper contextual relationships between features and learning styles.

Moreover, enhanced strategies for managing class imbalance should be investigated. Potential solutions include hybrid resampling approaches, cost-sensitive learning models, and adaptive loss functions that prioritize underrepresented classes without compromising overall accuracy. The development of models capable of self-monitoring and recalibrating their learning objectives based on observed performance could further reduce dependency on any single class.

Lastly, incorporating explainability techniques, such as SHAP or LIME, would provide valuable insights into the model's decision-making process, increasing trust and transparency in educational applications. This would facilitate stakeholder engagement, particularly among educators and curriculum designers, and promote the practical adoption of such systems in personalized learning environments.

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## Appendix

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### 5.4 Mapping of Course and Program Outcomes

#### 5.4.1 CSE400-A

##### **PO1 (Engineering Knowledge)**

At this stage, our work focused on identifying a meaningful and feasible problem in the domain of education technology that could be addressed using engineering knowledge from the Computer Science and Engineering field. After reviewing various real-world challenges, we selected the problem of automatic learning style detection in e-learning environments. Our decision was based on the growing demand for personalized education and the limitations of traditional self-assessment-based methods like the ILS questionnaire. The topic aligns with concepts learned in machine learning, data science, and artificial intelligence. Though no model was built during this phase, we developed a clear understanding of the theoretical background necessary for implementation in CSE400-B.

##### **PO4 (Investigation)**

We conducted an extensive investigation into the field of learning analytics by reviewing a wide range of academic journals, IEEE papers, and research articles. This literature review helped us understand the different approaches researchers have taken to detect learning styles, the role of the Felder-Silverman Learning Style Model (FSLSM), the use of LMS behavior data, and the common machine learning techniques applied. It also revealed limitations in existing work, such as the over-reliance on manual surveys and a lack of generalizability. These insights shaped our own research direction and helped us identify a gap that our capstone project could address.

## CO1: Problem Exploration and Identification

| CO  | Details                                                                                                                                                                                                                                       | Knowledge Profile (K)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Engineering Problem (EP)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO1 | We explored and selected a real-life problem in the education domain: detecting learners' cognitive styles automatically using LMS interaction data. This problem, although yet to be implemented, is complex and rooted in real-world needs. | <p><b>K1:</b> Understanding of natural sciences and foundational theories, including learning theories, behavioral data, and classification problems.</p> <p><b>K2:</b> Application of mathematical/statistical reasoning in evaluating algorithms like Decision Trees, XGBoost.</p> <p><b>K3:</b> Core engineering knowledge such as model selection, data handling, evaluation metrics.</p> <p><b>K4:</b> Specialist knowledge in machine learning, educational data mining, and personalized learning frameworks.</p> | <p><b>EP1:</b> Deep conceptual understanding used to identify a solvable, impactful problem.</p> <p><b>EP2:</b> Consideration of conflicting requirements (e.g., accuracy vs. interpretability).</p> <p><b>EP3:</b> Absence of off-the-shelf solutions guided us toward original implementation.</p> <p><b>EP4:</b> Acknowledgment of real-world constraints like LMS data variability and generalizability.</p> <p><b>EP5:</b> Ensured the problem matched engineering-level complexity and feasibility.</p> <p><b>EP6:</b> Accounted for future users and contextual adaptability in education systems.</p> |

## CO2: Problem Definition and Scoping

| CO  | Details                                                                                                                                                                                                                                          | Knowledge Profile (K)                                                                                                                                             | Engineering Problem (EP)                                                                                                                                                      |
|-----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO2 | Through problem scoping and literature review, we defined the challenge of improving personalized learning by eliminating manual learning style assessments. Our project aims to develop an automated, behavior-driven classification framework. | <p><b>K8:</b> Comprehensive literature review across FSLSM, educational machine learning, SMOTE, ensemble algorithms, and behavioral modeling using LMS data.</p> | <p><b>EP1–EP6:</b> (Same as CO1)</p> <p><b>EP7:</b> Refinement of the problem based on feasibility, stakeholder needs, and the potential impact on educational practices.</p> |



## 5.4.2 CSE400-B

### CO3: Problem Analysis and Modeling

| CO  | Details                                                                                                                                                                                                                                                                                        | Knowledge Profile (K)                                                                                                                                                                                                                                                                                                                                                                                           | Engineering Problem (EP)                                                                                                                                                                                                                                                                                                                                                                         |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO3 | We analyzed various components of the proposed problem to design a machine learning-based solution. This included studying the FSLSM framework, extracting key features from the dataset, understanding the imbalance in class distribution, and selecting suitable classification algorithms. | <b>K1:</b> Reviewed behavioral and cognitive science theories to understand learning preferences.<br><b>K2:</b> Applied statistical analysis and performance metrics like accuracy, precision, recall, and F1-score.<br><b>K3:</b> Studied and implemented ML algorithms (Random Forest, Decision Tree, SVM, XGBoost).<br><b>K4:</b> Utilized tools such as scikit-learn, XGBoost, SMOTE, and imbalanced-learn. | <b>EP1:</b> Decomposed the problem across FSLSM dimensions.<br><b>EP2:</b> Managed conflicting factors like imbalance and feature correlation.<br><b>EP3:</b> Iterative design was needed due to lack of existing implementation.<br><b>EP6:</b> Addressed real-world usability in solution design.<br><b>EP7:</b> Tackled interdependent subtasks like SMOTE balancing, tuning, and evaluation. |

### CO4: System Design and Implementation

| CO  | Details                                                                                                                                                                                                          | Knowledge Profile (K)                                                                                                                                          | Engineering Problem (EP)                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO4 | We designed and developed a modular classification system capable of identifying learning styles from behavioral data. The system includes preprocessing, SMOTE-based balancing, model training, and evaluation. | <b>K5:</b> Built complete ML pipelines using Python (pandas, scikit-learn, matplotlib, seaborn, XGBoost). Applied hyperparameter tuning and k-fold validation. | <b>EP1:</b> Designed a full pipeline combining preprocessing and model analysis.<br><b>EP2:</b> SMOTE integration and model tuning required staged design.<br><b>EP4:</b> Encountered challenges in feature selection and imbalance visualization.<br><b>EP5:</b> Ensured reproducibility and model robustness.<br><b>EP6:</b> System accounted for varied learner behaviors.<br><b>EP7:</b> Built modular components from data to decision logic. |

### CO5: Application of Modern Tools

| CO  | Details                                                                                                                                                                    | Knowledge Profile (K)                                                                                                  | Engineering Problem (EP)                                                                                                                                                                                                                                                                             |
|-----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO5 | Modern IT tools and platforms were used throughout the project lifecycle. We worked in Google Colab and used popular libraries for development, tuning, and visualization. | <b>K6:</b> Used pandas, numpy, matplotlib, scikit-learn, XGBoost, and imbalanced-learn for full ML pipeline execution. | <b>EP1:</b> Utilized open-source tools for end-to-end development.<br><b>EP2:</b> Fine-tuned models to align with SMOTE outputs.<br><b>EP4:</b> Managed tool limitations (e.g., long processing during validation).<br><b>EP5:</b> Designed output for clear reproducibility and future scalability. |

### 5.4.3 CSE400-C

#### CO6: Ethical, Societal, and Environmental Considerations

| CO  | Details                                                                                                                                                                                                                                                     | Knowledge Profile (K)                                                                                                                                                  | Engineering Problem (EP)                                                                                                                                                                                                                                                                    |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO6 | We carefully considered societal, ethical, and educational impacts of deploying an automated learning style detection system. The goal is to enhance student learning outcomes through personalization without introducing bias or compromising user trust. | <b>K7:</b> Gained awareness of the influence of machine learning systems on learner experiences and the need for responsible, interpretable, and fair AI in education. | <b>EP2:</b> Acknowledged ethical concerns, such as fairness and bias, particularly for underrepresented learning styles.<br><b>EP5:</b> Considered reliability and user trust in system predictions.<br><b>EP6:</b> Addressed diverse user expectations regarding AI-based personalization. |

#### Program Outcome (PO7): Environment and Sustainability

The proposed system promotes sustainability by replacing printed learning style questionnaires with digital, behavior-based assessments. This reduces paper use and supports a fully online, eco-friendly personalized learning framework.

#### Program Outcome (PO8): Ethics

Ethical considerations were integral to our project planning. Though we used behavioral data rather than sensitive personal records, we prioritized privacy and transparency. Future work includes integrating explainable AI (e.g., SHAP) to reduce potential bias and improve fairness.

#### Program Outcome (PO9): Individual and Team Work

Teamwork played a crucial role in the project's success. Each member contributed individually to exploring models, literature, and development tasks, while regular collaborative sessions ensured integration, validation, and refinement of the overall system.